# CHAPTER – 1 : NATURE AND SCOPE OF ECONOMICS

## UNIT I: INTRODUCTION

### 1.0.0 WHAT IS BUSINESS?

A business is an economic activity. The purpose of any economic activity is to earn profit. Therefore, the main target of any business is to maximize its profit.

(How to maximize profit will be seen in chapter 4).

### 1.0.1 WHAT IS ECONOMICS?

The term economics derived from a Greek word "Oikonomia", which means 'House Hold' till 19th century, economics was known as political economy.

As per Adam smith, economics deals with wealth and a book named "an inquiry into the nature & causes of the wealth of nations (1776)" is written by him & is considered as the first modern work of economics.

As per Robin economics is nothing but "choice making" because of the two fundamental facts:-

i. human being have unlimited wants and

ii. the means to satisfy these wants are relatively scarce.

The study of economics will enable us to develop an analytical approach that help us in understanding and analyzing a wide range of economic issues.

### 1.0.2 MEANING OF BUSINESS ECONOMICS

**(a)** Case study of worldwide food limited: (Refer ICAI SM).

**(b)** Management of any business have to take many strategic, tactical and operational decisions. These decision making is not simple but this effect our business in present & future.

Business economic help management to take good business decision through its practical approach business economic have close connection with micro as well as macro economics, and also it use concept of "operational research", statistics, mathematics and theory of decision making.

## 1.1 DEFINITIONS OF BUSINESS ECONOMICS

As per Joel dean Business economics use the economic analysis in the formulation of business policies and it is a part of applied economics.

## 1.2 NATURE OF BUSINESS ECONOMICS:

Business economics is based on micro economics but uses the concept of macro economics.

**Micro economics:** under micro economics we study the behavior of diff. individual or organization or company.

**Example:** Product prices, consumer behavior, factor price, *(all concept & study in chapter 2,3,&4 are covered under micro approach).*

**Macro economics:** Under macro economics we study the overall economics concept of country. Example: national income, balance of payment, external value of currency, unemployment etc.

### NATURE OF BUSINESS ECONOMICS-

1) **Based on micro economics:** because in business economics we take decisions of any business / firm or company not for any country.

2) **Business economics is a science:** because it have "cause and effect" relationship. (ex. Law of demand have cause & effect i.e. P ↑Q ↓).

3) **Incorporates elements of macro analysis:** all business are affected by external environment of whole economy, therefore while taking decision of any business we must incorporate macro analysis, like inflation, reporate, import-export policy, taxation etc.

4) **Business economics is an art:** it involves practical application of any decision taken.

5) **It uses the theory of market and private enterprise:** under this business eco. Always take decision as per present market situation and also for profit maximization.

6) **Pragmatic (practical) in approach:** because it tackles practical problems of any firm.

7) **Interdisciplinary in nature:** because it uses / incorporates tools from other disciplines such as maths, or, accounting etc.

8) **Normative in nature:** Business economics theory has developed along two lines positive and normative. Positive / pure science analyses cause & effect relationship and in simple wording it states what is (fact).
Normative science involves value judgement and it is prescriptive in nature and it suggest "what should be" a particular decisions.

9) Business economics is generally normative but it also required positive approach to create / judge any fact.

## 1.3 SCOPE OF BUSINESS ECONOMICS

Its scope is very wide and covers most of practical problems of any business. It includes following:

1. **Micro economics** applied to operational or internal issue.
2. **Macro economics** applied to environmental or external issue.

1. **Operational / Internal Issues**

(i) Demand analysis and forecasting.
(ii) Production and cost analysis.
(iii) Inventory management (Raw material, finished goods etc.) concept:- abc analysis.
(iv) Market structure and pricing policy.
(v) Resource allocation: by using linear programing.
(vi) Capital and investment decision: by using concept of capital budgeting.
(vii) Profit analysis.
(viii) Risk and uncertainty analysis

**2. Environment or External Issues**

(i) Type of economic system: (capitalistic / socialistic & mix economy.)

(ii) Business cycle (chapter. 5)

(iii) General trends in national income, employment, price.

(iv) Governments policies: ex. Taxation, Exim policy

(v) Socio – economic organization: trade / workers unions.

(vi) Political environment

(vii) Working of central banks, financial sector, capital market etc.

## 1.4 DIFFERENCE BETWEEN ECONOMICS & BUSINESS ECONOMICS

| Basis of Difference | Economics | Business Economics |
|---|---|---|
| Meaning | It involves the framing of economic principles to solve economic problems. | It involves the application of economic principles to solve business problems. |
| Character | It is microeconomic as well as macroeconomic in character. | It is microeconomic in character. |
| Main Task | The fulfilment of needs of individuals as well as entities. | Proper decision making inparticular business entity. |
| Nature | It is positive as well as normative in nature. | It is only normative in nature. |
| Scope | It has a wider scope. | It has a comparatively narrow scope. |
| Branches | It has business economics asits applied branch. | It is an applied branch of economics. |
| Concerned with | All the theories from production to consumption including distribution. | It is concerned with only profit theory ignoring other theories. |
| Analysis Involved | It includes the analysis of macro level issues like growth, inflation and employment, etc. | It includes the analysis of micro level issues like demand, supply and profit etc. |
| Concentration | It concentrates only on the economic aspects of any business problem. | It concentrates on both economic as well as non- economic aspects of any business problem. |
| Validity of Assumptions | It is based on certain assumptions. | Some assumptions become invalid when applied. |

"Study of Inventory Management, Product and Promotion Policy, Resource Allocation, Capital Budgeting, Risk and Uncertainty analysis are outside the scope of this book. They will be taught in other subjects – Financial Management, Strategic Management etc. at higher levels of CA course."

# CHAPTER – 1 : NATURE AND SCOPE OF ECONOMICS

## UNIT II: BASIC PROBLEMS OF AN ECONOMY & ROLE OF PRICE MECHANISM

All countries have "scarcity", it means its resources are limited with their object, therefore all countries have to make choice between alternative uses of limited resources.

While selecting alternative uses, four central economic problem / questions arises to all types of economic system (capitalistic / socialistic / mix).

**Following are four basic economic problems:**

a) What to produce?

b) How to produce?

c) For whom to produce?

d) What provisions are to be made for economic growth?

**Answers of above four economic question as per capitalistic / socialistic economy:-**

**a) What to produce?**

As per capitalistic system, those goods will be produce which is profitable and having good demand but socialistic system will produce goods which is good for public even if not profitable.

**b) How to produce?**

There are various alternative techniques of producing good. (ex. Manual vs. automatic) the capitalistic system will produce either through manual or automatic which is having less cost but social system will use manual if its economy is labour surplus economy and it will use automatic if its economy is capital surplus economy.

**c) For whom to produce?**

A society can not satisfied each and every want of all people. Under capitalistic system production will be for those who can pay for it (having purchasing power) but under socialistic system produce for those who have need for this.

**d) What provision should be made for economic growth?**

A society would not like to use all its scarce resources for current consumption only under capitalistic system the companies is not interested in a project having long term

investment but under social system the govt. is interested for it.

## 2.1 CAPITALIST ECONOMY

(say free market economy, "laissez faire economy")

Following are the characteristics of capitalist economy :-

1. **Right to property:** under this economy every individual person / company can purchase land/ building etc. and create its ownership, however government can put some restrictions on it.

2. **Freedom of enterprise / company :** a producer is free to produce any goods and also free to set any price subject to some restriction in some industry by govt.

3. **Freedom of economic choice :** freedom to consumer to purchase any goods.

4. **Profit motive:** company always work for maximum profit.

5. **Consumer sovereignty (consumer is king) :**

6. **Competition :**

7. **Absence of govt. interference :**

## 2.1.0 HOW DO CAPITALIST ECONOMICS SOLVE THEIR CENTRAL PROBLEM

## 2.1.1 MERITS OF CAPITALIST ECONOMY:

1. Capitalism is self regulating & works through price mechanism.
2. Due to right of private property and profit motive the company have driving force to work more.
3. Economic growth is faster because investors try to invest in only profitable and feasible project.
4. Optimum allocation of resources.
5. **Operative efficiency :** therefore cost of production will minimize.
6. Offer incentives for efficient decision & implementation.
7. Higher living standard.
8. Innovation & technological progress.
9. Function in democratic framework.

10. This system encourages person for taking risk and become businessman.

## 2.1.2 DEMERITS OF CAPITALISM

1. **Inequality and social injustice** : this system split society into two classes normally "haves and have not's" and their is huge gap between rich & poor.
2. **Property right over human rights**
3. **Wide differences in economic opportunities.**
4. **Ignores human welfare.**
5. **Due to inequalities the market demand may not be real needs of society.**
6. **Exploitation of labour.**
7. **Consumer sovereignty is a myth:** due to advertisement.
8. **Misallocation of resources into luxury goods**
9. **Produce harmful good due [huge demand].** As they are more profitable
10. **Unemployment:** more use of capital intensive technique.
11. **Monopoly power**
12. **Environment degradation**

## 2.2 SOCIALIST ECONOMY

This concept Is given by *karl mark and Frederic engels in their work "The communist manifesto" published in 1848 under this economy* all production units are owned by govt. this economy may be called command economy or centrally planned economy. In this system there is no role of price mechanism and market force. Under socialist economy, production and distribution of goods are aimed to maximize welfare of public.

**Characteristics of social economy**

i. Collective ownership.
ii. Economic planning.
iii. Absence of consumer choice.
iv. Relatively equal distribution.
v. Minimum role of price mechanism
vi. Absence of competition.

## 2.2.0 MERITS OF SOCIALISM (Read with ICAI SM)

## 2.2.1 DEMERITS OF SOCIALISM (Read with ICAI SM)

## 2.3 MIX ECONOMY:

The mix economy system depends on both market and govt. for allocation of resource in fact every economy in the real world makes use of mix economy. The aim of this economy to use best features of both economy.

## 2.3.0 FEATURES OF MIX ECONOMY:

Co- existence of private and public sector (private sector / public sector / combined sector).

**Merits:**

1. Economic freedom with planning.
2. Price Mechanism.
3. Consumer sovereignty / freedom of choice.
4. Incentive for innovation.
5. Encourage risk taking.
6. Social equality.

**Demerits:**

1. Mixed economy, sometimes, is characterized by excessive controls by the state resulting in reduced incentives and constrained growth of the private sector, poor implementation of planning, higher rates of taxation, lack of efficiency, corruption, wastage of resources, undue delays in economic decisions and poor performance of the public sector.

2. Moreover, it is very difficult to maintain a proper balance between the public and private sectors.

3. In the absence of strong governmental initiatives, the private sector is likely to grow disproportionately. The system would then resemble capitalism with all its disadvantages.

# CHAPTER – 2 : THEORY OF DEMAND & SUPPLY

## UNIT I: LAW OF DEMAND AND ELASTICITY OF DEMAND

### 1.0 MEANING OF DEMAND

"The term demand refers to the quantity of goods or services that buyers are "willing and able" to purchase at various prices during a given period of time".

Demand = WILLINGNESS and ABLENESS.

↓ ↓

It creates due to "utility" it depend on income.

↓

Utility is "want satisfying power" of any goods.

For effective demand there should be following 3 condition:

(a) Willingness (ie. desire)

(b) Ableness (ie. Means to purchase / money)

(c) Willingness to use those means (money) for that purchase.

### 1.1 WHAT DETERMINES DEMAND?

The factors which are responsible for increase or decrease in demand

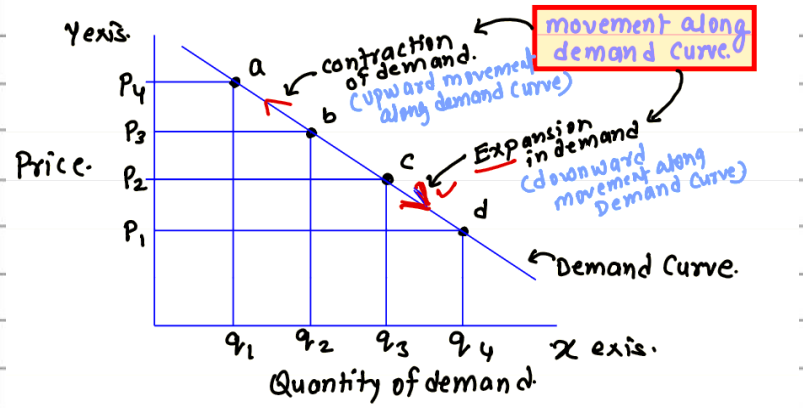
**(i) Price of the commodity :**

P ↓ → D ↑ (Expansion of demand)

P ↑ → D ↓ (Contraction of demand)

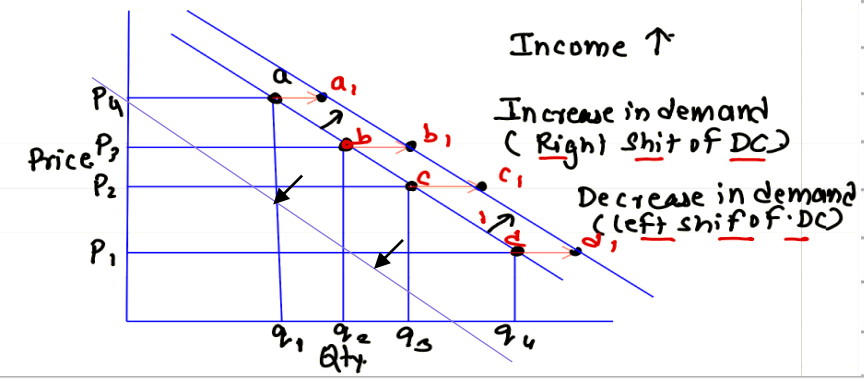
**NOTE:**

(1) Expansion / contraction of demand may be called increase in quantity / decrease in quantity of demand.



(2) If any change in demand due to other factors (other the price), then we will called. Increase / decrease in demand and the demand curve will shift to right / left.

**Example:** Income ↑ D ↑ (increase in demand)
Income ↓ D ↓ (decrease in demand)




**(ii) Price of related commodity**

(a) Coffee D↑ (Increase) Vs Tea P↑

Positive Relation (Direct Relation) between price of Substitute goods (Tea) and demand of main goods (coffee).

(b) Suger D↓ (Decrease) Vs. Tea P↑

Negative Relation between price of complimentary goods and demand of ~~our~~ main goods (Suger)

**(iii) Disposable income of consumer**

**(a) General goods**

Income ↑ ↓ Demand ↑ ↓

**(b) Inferior goods**

Income ↑ ↓ Demand ↓ ↑

(c) **Essential consumer goods**

Income ↑ 50% Demand ↑ 5%

**Note:** Business manager should recognize the movement in macro-economic factors which effect the buyer's income, so that they can forecast demand.

(iv) **TASTES AND PREFERENCES OF BUYER :**

(a) **Demonstration effect :-** consumer like what he see.

(b) **Bandwagon effect :-** To be fashionable or stylish.

(c) **Snob effect :-** To be exclusive.

(d) **Veblen effect:-** expensive goods have more demand amongst rich person.

(v) **CONSUMER EXPECTATION:**

Consumer expectation regarding future –

(a) **Price** ↑ **current demand** ↑

(b) **Income** ↑ **current demand** ↑

(c) **supply** ↓ **current demand** ↑

(vi) **OTHER FACTORS :**

(a) Size of population : population ↑ Demand ↑

(b) Age distribution of population :

**Example:** If more children in population then demand of toys will increase.

(c) National income and its distribution :

(i) National income ↑ Demand ↑

(ii) Distribution of national income-

If even D ↑

If uneven D ↓

**Note:** propensity to consume will be less in rich person and high in poor person.

(d) Consumer credit facility and interest rate :

Good & easy loan facility : Demand ↑

Low interest rate : Demand ↑

**Example:** Demand of car in India is high because of easy & low interest car loan

(e) Govt. policy:

Taxation ↑ demand ↓

Subsidy ↑ demand ↑

Govt. purchase ↑ demand ↑

## 1.2 DEMAND FUNCTION.

Demand function is a "symbolic statement" of a relationship between determinants of demand (independent variable) and demand (dependent variable).

Demand function : $Q_x = f(P_x, Y, P_r.....)$

Where $Q_x$ is the quantity demanded of product X

$P_x$ is the price of the commodity

Y is the money income of the consumer, and

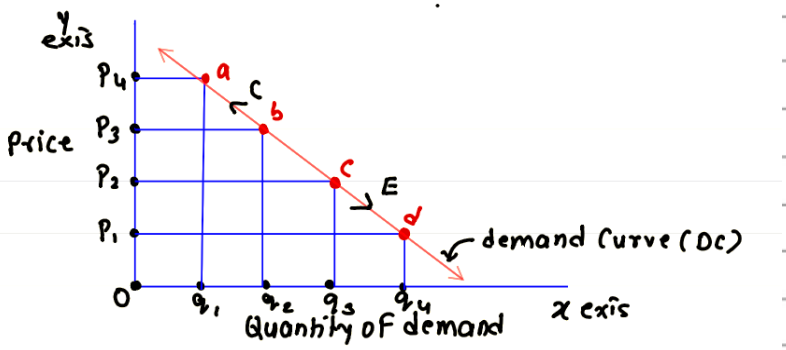
$P_r$ is the price of related goods

## 1.3. THE LAW OF DEMAND.

**Law of demand as per Prof. Alfred Marshall :-** "The greater the amount to be sold, the smaller must be the price at which it is offered in order that it may find purchasers or in other words the amount demanded increases with a fall in price and diminishes with a rise in price".

In simple wording law of demand is if "other factors" are constant (i.e. not changing / ceteris paribus) then if price increase the quantity of demand will decrease (i.e. constraction) and vice versa.

**Demand curve and law of demand:**




**NOTE :-**

1. DC is downward sloping to right or we can say negative slope.
2. DC is negative slope because there is inverse relation between price & qty of demand.
3. If we are moving from point A to D then it is expansion of demand (downward movement along DC)

4. If we are moving from point D to A then it is contraction of demand (upward movement along DC)
5. 'Movement along DC', E&C.
6. Law of demand : if other factors are constant. Then this movement along demand curve is nothing but law of demand.
7. Ceteris paribus : it means other factors are constant.

## 1.3.0 THE DEMAND SCHEDULE

A table showing the quantity of a goods that buyers would choose to purchase. At different price per unit of time.

| S.NO. | PRICE (₹) | QTY. DEMAND (UNITS) |
|---|---|---|
| 1 | 10 | 100 |
| 2 | 15 | 80 |
| 3 | 20 | 50 |

## 1.3.1 DEMAND CURVE

A graphical presentation of demand schedule.

## 1.3.2 MARKET DEMAND SCHEDULE

Table showing the quantity demanded by all the buyers in the market at different price per unit of time.

"MARKET DEMAND SCHEDULE"

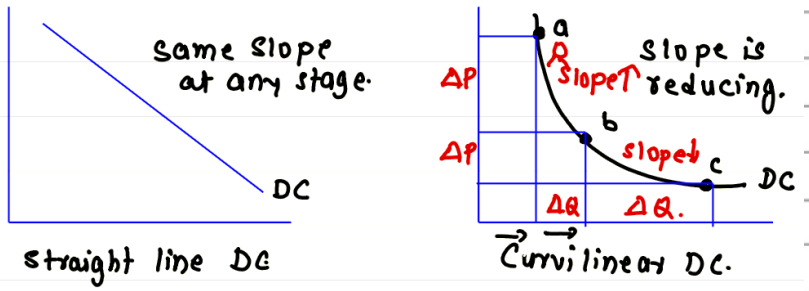
| S.NO. | PRICE (₹) | MR. A | MR. B | MR. C | MARKET DEMAND |
|---|---|---|---|---|---|
| 1 | 5 | 70 | 50 | 20 | 140 |
| 2 | 3 | 80 | 60 | 30 | 160 |
| 3 | 2 | 100 | 70 | 40 | 210 |

**Note:** market demand schedule is horizontal / lateral summation of individual demand of each buyer.

## 1.3.3 MARKET DEMAND CURVE

A graphical presentation of market demand schedule.

Straight line DC vs. Curvilinear DC.




**Note: - 1)** Slope of demand curve = $\Delta P / \Delta Q$

2) Straight line demand curve is not practical but "convenient tools for analysis."

## 1.3.4 RATIONAL OF THE LAW OF DEMAND:

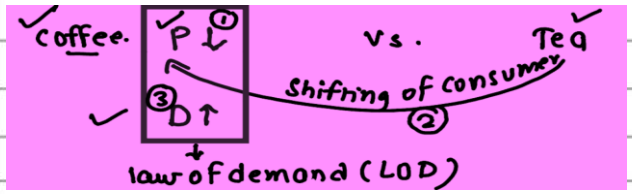
(Reasons / logics of law of demand)

1. **Price effect of fall In price-**

**Law of demand**
If other factors are constant
P ↑ D ↓
P ↓ D ↑

(a) **Substitution effect :**

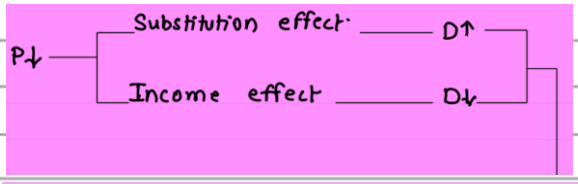
**Example:**



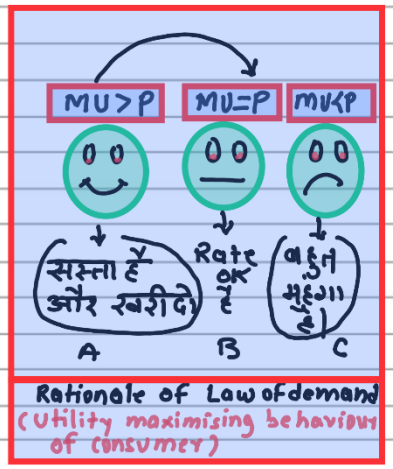

**NOTE :** The substitution effect will be stronger when: i. close substitute ii. lower cost of switching iii. lowest inconvenience.

(b) **Income effect:** when price decrease, we can buy more goods with same income, it means our income become more effective or we can say our purchasing power increase or we can also say our real income increase.

(c) **Price effect on inferior goods.**




"The law of demand will apply only if substitution effect is more powerful then income effect".




2. **Utility maximizing behavior of consumer-**

A consumer always try to maximize his satisfaction. For this he always maintain "Price (P) = Marginal utility (MU)". Therefore, if P < MU, then he increase the quantity purchase till P = MU and if price again reduces (i.e. P < MU) then again in increase quantity of demand therefore we conclude that if P ↓ D ↑.

3. **Arrival of new consumer-**

When the price of a commodity falls, more consumers start buying it because some of those who could not afford to buy it earlier may now be able to buy it.

4. **Different uses-**

Many commodities have multiple uses.

## 1.3.5 EXCEPTIONS OF LAW OF DEMAND

### WHAT IS LAW OF DEMAND?

"if other factors are constant then P ↑ D ↓ or P ↓ D ↑"

Exception of law of demand :- (i.e. P ↑ D ↑ or P ↓ D ↓ )

**(i) Conspicuous goods :**

- Example : Dimond
- If P ↑ then Dimond become more attractive amongst rich person
  ∴ D ↑
- It may be called Veblen effect or prestige goods effect.

**(ii) Giffen goods:**

- This concept given by "sir Robert Giffen" (Scottish economist)
- Giffen goods = [inferior goods + no close substitute + substantial part of consumer income]

**Conclusion:** all Giffen goods are inferior goods. But all inferior goods are not Giffen goods.

**Giffen Effect:**

**Example:** Bread is giffen goods vs cake is normal good

| Price of bread | No. of days of bread consume | No. of days of cake consume |
|---|---|---|
| 10 | 25 days | 5 days |
| 15 | 26 days | 4 days |
| 20 | 27 days | 3 days |
| 25 | 28 days | 2 days |

In the above example price of the bread increase and also the demand of bread increases. So this is an exception of law of demand in case of giffen goods (i.e. bread).

**(iii) Conspicuous necessities:**

If any good, become the part of our life, then even if its price increases the consumers will not reduce its demand but they increase its demand. For example- Mobile, TV, Car etc.

**(iv) Future expectation about prices.**

If future expectation of price increase : ↑

current price ↑ current demand ↑

**Example:-** share market.

(v) **Incomplete information and irrational behavior of consumer:-**

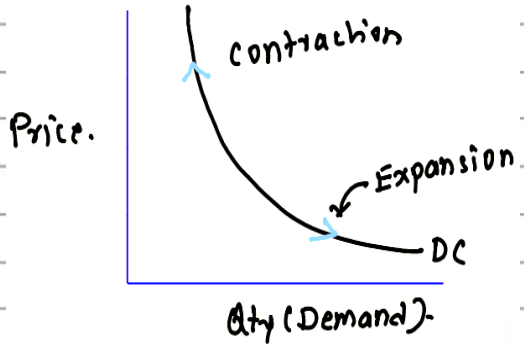
- No information of market or current / past price. Then consumer can buy more at higher price.
- If consumer is not calculating the usefulness/ utility of a product.

(vi) **Demand for necessaries:** the law of demand does not apply much in the case of necessaries of life. i.e. medicine, salt.

(vii) **Speculative goods:** in the speculative market, particularly in the market for stock and shares, more will be demanded when the prices are rising and less will be demanded when price decline.

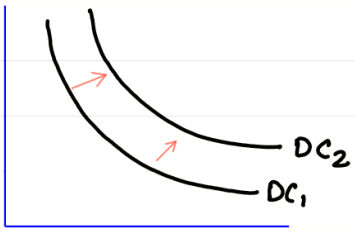
## 1.4 EXPANSION AND CONTRACTION OF DEMAND (E/C)

a) E/C is due to change in price.
b) E/C may be called increase / decrease in quantity demand
c) E/C is movement along demand curve
d) Expansion = downward movement in DC.
e) Contraction = upward movement in DC.

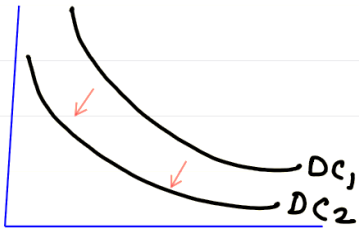



### 1.4.1 Increase and decrease in demand

a) I/D in demand is due to other factors (non- price factors.)
b) Increase in demand = right shift of DC.
c) Decrease in demand = left shift of DC.




Right shift (Increase)




Left shift (decrease.

## 1.4.2 USE OF ABOVE CONCEPT IN BUSINESS DECISION. (E/C VS I/D)

(a) When demand increase due to other factors it means firm can sell more at the existing price.

(b) By advertisement a firm can shift DC rightward.

(c) Additional demand is not free of cost.

## 1.5.1 ELASTICITY OF DEMAND

1) "It is responsiveness of demand due to change in any factor."

2) If nothing is given then price elasticity of demand will be considered.

3) Price elasticity of demand [PED] Ep

**Formula:**

**(a) Percentage method**

$$Ep = \frac{\text{Percentage change in quantity demanded}}{\text{percentage change in price}}$$

OR

$$Ep = \frac{\Delta q}{\Delta p} \times \frac{p}{q}$$

(where p = old price & q = old quantity)

**(b) Mid-point method (Arc method)**

$$E_p = \frac{q_2 - q_1}{q_2 + q_1} \times \frac{p_2 + p_1}{p_2 - p_1}$$

(where, $p_1$ = old price, $p_2$ = New price, $q_1$ = Old quantity & $q_2$ = New quantity)

**Example 1.**

If price increase by 10%, the quantity of demand decrease by 5 % then Ep = ?

**Answer: 0.5**

**Example 2.**

| | Price (₹) | Quantity demand (units) |
|---|---|---|
| Old | 15 | 1000 units |
| New | 20 | 700 units |
| Ep = ? | | |

**Answer: (-) 0.9**

**Example 3.**

Price decrease from Rs. 6 to Rs. 4 and therefore. Quantity of demand increase from 10 units to 15 units. Find the co-efficient of price elasticity.

**Hint:** if we ask for "co-efficient" of price elasticity then we have to present answer "without sign" ( +/ -).

**Answer: 1.5.**

**Example 4.**

If price decrease from Rs. 100 to Rs. 60 and the Ep = 1.5 and the original (old) Qty is 30 units, then find the new quantity demanded.

**Answer: 48.**

**Example 5:** A consumer buy 80 units of goods at the price of Rs. 4 per unit, suppose the price elasticity is (-) 4, then at what price he will buy 60 units?

**Answer: 4.25**

| **Example 6:** | Old | New |
|---|---|---|
| Price | 100 | 150 |
| Quantity | 500 | 300 |

Find the price elasticity as per percentage method and also as per mid point method.

**Answer: (-)0.8 / (-) 1.25**

## 1.5.2 POINT ELASTICITY

(i) Normally in percentage method or arc method we calculate elasticity between two price & two quantity or we can say between two point on a demand curve.

(ii) Point elasticity is being calculated on a single point of straight line demand curve.

(iii) Formula

$$Ep = \frac{-dq}{dp} \times \frac{p}{q}$$

Where "dq/dp" is derivative of quantity with respect to price

**Example:**

Find the point elasticity of demand at price of Rs. 8/- if demand function is $Q_d$ = 1200-100P

**Solution:**

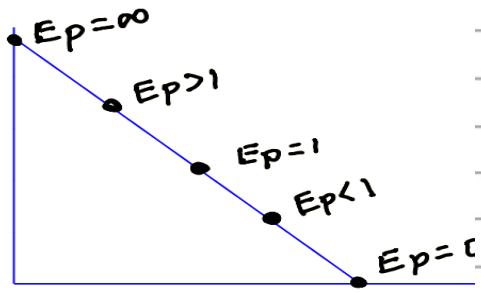
$$Ep = \frac{-dq}{dp} \times \frac{p}{q}$$

= 100 x 8/400

= (-) 2

(iv) Point elasticity – "Geometric method"

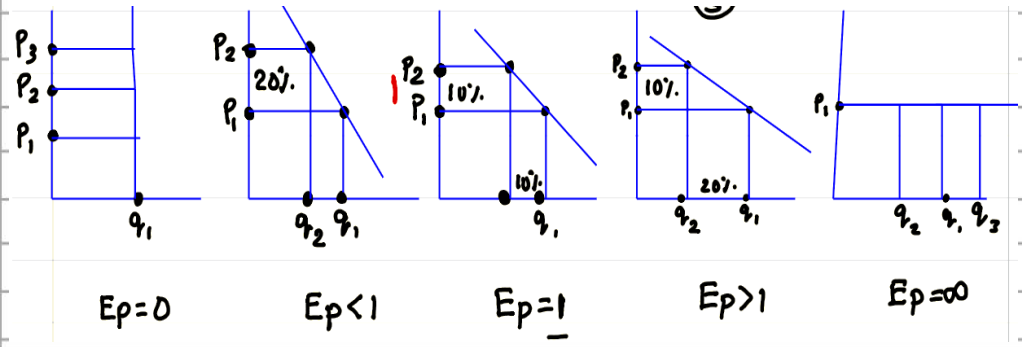
$Ep = \frac{\text{Lower segment}}{\text{Upper segment}}$




## 1.5.3 INTERPRETATION OF NUMERICAL VALUE OF ELASTICITY OF DEMAND

| S. NO. | VALUE | TERMINOLOGY |
|---|---|---|
| 1. | Ep = 0 | Perfectly inelastic |
| 2. | Ep < 1 | Inelastic (relatively less elastic) |
| 3. | Ep = 1 | Unit elastic |
| 4. | Ep > 1 | Elastic (relatively more elastic) |
| 5. | Ep = ∞ | Perfectly elastic |

**Demand curve under five elasticity**

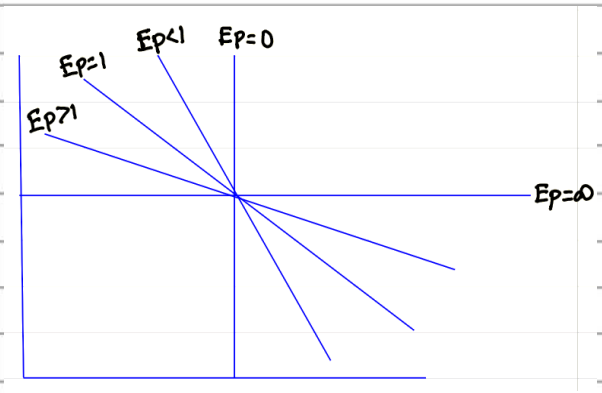



**Note:**

1. When elasticity increases (i.e. Ep = 0 to Ep = ∞) the demand curve become flatter from stepper and visa-versa.
2. If we move from Ep = 0 to Ep = ∞, the demand curve will convert from vertical to horizontal anti-clockwise.
3. $Ep = \frac{\Delta q}{\Delta p}$ v/s slope = $\frac{\Delta p}{\Delta q}$

When the Ep increasing, the slope will decreases.

4. Consolidated demand curve of all elasticity.




## 1.5.4 TOTAL OUTLAY METHOD OF CALCULATING PRICE ELASTICITY

**[TOTAL EXPENDITURE METHOD]**

(a) **Total expenditure (TE) = Price (P) x Quantity (Q)**

(b) **Total revenue (TR) = Price (P) x Quantity (Q)**

(c) **TE = TR**

**Example:** If a consumer (Mr. B) purchase 70 units of a goods at the price of Rs. 9 per unit from seller (Mr. S) then what will be the TE for Mr. B and also TR for Mr. S.

**Answer: 630.**

**Finding elasticity through change in price and effect on TE/TR**

| Cause 1. | Effect | | | Conclusion |
|---|---|---|---|---|
| P ↑ | TE↓ | or | TR↓ | Ep > 1 |
| P ↓ | TE↑ | or | TR↑ | |

| Cause 2. | Effect | | | Conclusion |
|---|---|---|---|---|
| P ↑ | TE↑ | or | TR ↑ | Ep < 1 |
| P ↓ | TE↓ | or | TR ↓ | |

| Cause 3. | Effect | Conclusion |
|---|---|---|
| P ↑ | TE =/ TR = (no change) | Ep = 1 |
| P ↓ | TE = / TR = (no change) | Ep = 1 |

**Note :-**

1) The main drawback of this method is that we can not calculate the value (co-efficient.) of elasticity.

2) By this method a business firm make policy to increase their "TR", for this they increase price of product having Ep < 1 and decrease the price of product having Ep > 1.

**Total Outlay/Total Expenditure Method For Elasticity Of Demand**

| Case | Price | TE = P x Q | TR = P x Q | Ep | p1 | q1 | $TE_1$ | p2 | q2 | $TE_2$ | Ep(arc) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Increase | decrease | decrease | E>1 | 10 | 100 | 1000 | 11 | 81.82 | 900 | -2.10 |
| 2 | Decrease | Increase | Increase | E>1 | 10 | 100 | 1000 | 9 | 116.67 | 1050 | -1.46 |
| 3 | Increase | Increase | Increase | E<1 | 10 | 100 | 1000 | 11 | 95.45 | 1050 | -0.49 |
| 4 | Decrease | decrease | decrease | E<1 | 10 | 100 | 1000 | 9 | 105.56 | 950 | -0.51 |
| 5 | Increase | No change | No change | E=1 | 10 | 100 | 1000 | 11 | 90.91 | 1000 | -1.00 |
| 6 | Decrease | No change | No change | E=1 | 10 | 100 | 1000 | 9 | 111.11 | 1000 | -1.00 |

**Note:** Formula used for elasticity:

$Ep = (q2-q1)/(q2+q1) \times (P2+P1)/(P2-P1)$.

## 1.5.5 Total Revenue [TR]

TR = Price x Quantity sold

(1) If supplier increase the price (P ↑)

Then price effect TR ↑

&

Quantity effect TR ↓

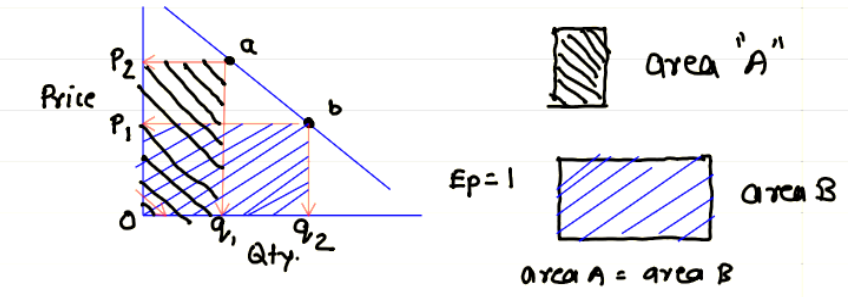
∴ If quantity effect is stronger [Ep>1] then TR ↓

&

If price effect is stronger [Ep <1] then TR ↑

(2) In case of Ep = 1 (Unit elasticity)

(i) P ↑ or P ↓ = TR will not change.




$TRp_1 = p_1 \times q_2$ [0 $p_1$ b$q_2$]

$TRp_2 = p_2 \times q_1$ (o$p_2$ a $q_1$)

Same ($TRp_1 = TRp_2$) as Ep = 1

(ii) In this situation a supplier can not generate extra revenue (i.e. marginal revenue) by changing price.
Marginal revenue (MR) = $TRp_1 - TRp_2$ = ZERO

(iii) When Ep = 1 then the shape of demand curve will be linear and the area of any point of demand curve will be same and this situation is called rectangular hyperbola.

## 1.5.6 Determinants of price elasticity of demand

**[Reasons of high (Ep>1) or low (Ep<1) price elasticity].**

**(1) Availability of substitute**

**(a)** A Goods having more substitute will have high elasticity (Ep>1 and a good having less / no substitute will have low elasticity (Ep<1).

**(b)** Generic goods vs branded goods
(Ep<1) (Ep>1)
i.e. salt i.e. Tata salt

**(2) Position of a goods in consumer budget**

**a)** More share in budget Ep>1
(Example: If we spend 40% of our income in good 'A') Ex. Food.

**b)** Less share In budget Ep<1
(Example: We spend only 1% of our income in good 'B') Ex. Salt/ newspaper

**(3) Nature of goods**

**a)** Important goods/ a goods for which we can not postpone our demand ⟶ Ep<1

**b)** Not a important goods / luxury goods / we can postpone demand of a goods ⟶ Ep>1

**(4) Number of uses :**

**a)** If any product having multiple uses (ex. Milk/electricity) Ep >1

**b)** Less use/ single use ⟶ Ep<1

**(5) Time to decide :**

**a)** If less time to decide to purchase any goods- Ep<1

**b)** If more time to decide to purchase any goods- Ep>1

**(6) Consumer habit :**

If consumer is habituated to any product ⟶ Ep<1

(ex. Gutka, Bidi, etc.)

**(7) Tied demand :**

| Tied demand | vs | Autonomous demand |
|---|---|---|
| (Ex. Ink cartridges of a particular brand of printer) ↓ Ep<1 | | (Ex. Printer) ↓ Ep>1 |

**(8) Price range :**

very low / very high price range goods ⟶ Ep<1

↓ ↓

(Pen of Rs. 5) (Dimond of Rs. 5 crore)

**(9) Minor complementary items :**

Petrol is minor complementary for luxury car ⟶ Ep<1

## 1.6 INCOME ELASTICITY OF DEMAND

$$Ei = \frac{\%\Delta \text{ in Qty of demand}}{\%\Delta \text{ in income}} \quad \text{or} \quad Ei = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

where y = original income

Q = original Qty demanded.

**Conclusions:**

(a) Normally "Ei" is positive but it will be negative in case of Inferior goods.

(b) Ei of luxury goods is positive and Ei > 1.

(c) Ei of necessities is positive and Ei < 1.

(d) Ei in long run is higher then short run.

**Example:**

If income of Mr. Y increase from Rs. 40,000 to Rs. 70,000 and the demand of goods "A" (normal goods) increase from 500 units to 600 units and goods 'B' (luxury goods) increase from 100 units to 180 units and goods 'C' (inferior goods) decrease from 200 units to 150 units. Compute income elasticity of goods A, B, C.

**Answer:** 0.27, 1.07, (-) 0.33.

## 1.7 CROSS PRICE ELASTICITY OF DEMAND

$Ec = \frac{\%\Delta \text{ in Qty of goods x}}{\%\Delta \text{ in price of goods y}}$ or $Ec = \frac{\Delta Qx}{\Delta Py} \times \frac{Py}{Qx}$

**Conclusion:** if goods 'x' & 'y' are

a) (i) Substitute ⟶ Ec is positive & less then one.

(ii) Close substitute ⟶ Ec is positive & more then one.

(iii) Perfect substitute ⟶ Ec = ∞

b) Complementary ⟶ Ec is negative

c) unrelated ⟶ Ec = 0

**Example:** Find the cross elasticity of product 'x' if change in demand due to change in price of following goods are as follows:-

a) Product 'A' p ↑ 10% product 'x' demand 5% ↑

b) Product 'B' p ↑ 10% product 'x' demand 15% ↑

c) Product 'c' p ↑ 0% ↑ / product x demand 5 % ↑

d) Product 'd' p ↑ 10% product x demand ↓5 %

e) Product 'e' p ↑10% product x demand 0 %

**Solution.**

a) Ec = 5% / 10% = + 0.5 substitute.

b) Ec = 15% / 10% = (+) 1.5 close substitute.

c) Ec = 5 % / 0% = ∞ perfect substitute.

d) Ec = 5% / 10% = (-)0.5 complementary.

e) Ec = 0% / 10% = 0 unrelated

## 1.8 ADVERTISMENT ELASTICTIY

$Ea = \frac{\%\Delta \text{ Qty of demand}}{\%\Delta \text{ in spending of advertisement expenditure}}$

$Ea = \frac{\Delta Q}{\Delta A} \times \frac{A}{Q}$ where. A = original adv. Exp.

Q = original Qty.

Example- Adv. Exp. 25,00,000 ⟶ 30,00,000

Demand 10,000 ⟶ 15,000

$Ea = \frac{\Delta Q}{\Delta A} \times \frac{A}{Q} = \frac{5000}{5,00,000} \times \frac{25,00,000}{10,000} = \frac{125}{50} = (+)2.5$

**Conclusion:**

If Ea>1 then we are getting more from advertisement and we have good market team but if Ea<1 then we are not getting good result from advertisement. if Ea = 0 then our adv. Exp. is just a waste of money.

# CHAPTER – 2 : THEORY OF DEMAND & SUPPLY

## UNIT II: THEORY OF CONSUMER BEHAVIOUR

**What we learn in this Chapter:**

(1) **What is utility?**

(2) **Marshall's law:** Law of diminishing marginal utility (MU) and concept of consumer surplus and consumer equilibrium.

(3) **Hicks & Allen's law:** Indifference curve (IC) analysis and concept of consumer equilibrium

### 2.0 NATURE OF HUMAN WANTS:

(1) Wants are unlimited and means to satisfied those wants are limited.

(2) We need to make choices due to point (1) above.

(3) All want can not be satisfied at once, but each want is satiable.

(4) Wants are competitive / complementary.

(5) Wants are subjective and it is different for different person.

(6) Wants are affected by income, taste, fashion etc.

### 2.1 CLASSIFICATION OF WANT

(1) **Necessaries**

(2) **Comforts**

(3) **Luxuries**

### 2.2 WHAT IS UTILITY?

"Utility is want satisfying power of any goods & services."

### 2.3 MARSHALL'S MARGINAL UTILITY ANALYSIS:

| Tea/ cup | Utility | |
|---|---|---|
| | Marginal utility (MU) | Total utility (TU) |
| 1 | 50 | 50 |
| 2 | 30 | 80 |
| 3 | 10 | 90 |
| 4 | 0 | 90 |
| 5 | -10 | 80 |

Total utility (TU) = $MU_1 + MU_2 + MU_3$ .....$MU_n$

(in above table n = 5)

Marginal utility (MU) = $TU_n - TU_{n-1}$ or $\Delta$ TU / $\Delta$ Q.

| Example: | Units consume | TU |
|---|---|---|
| | 10 | 500 |
| | 11 | 600 |

MU of 11th unit?

| Example: | Unit Consume | TU |
|---|---|---|
| | 10 unit | 700 |
| | 13 unit | 1000 |

MU of 13th units?

## 2.3.1 ASSUMPTIONS OF MARGINAL UTILITY ANALYSIS (MARSHALL'S)

(1) **Rationality:** "Consumer will try to maximize his satisfaction with limited money income."
To maximize satisfaction, consumer increase purchase having MU > P and increase its purchase upto a situation where MU = P.

(2) **Cardinal measurability of utility:** In simple words consumer can quantify his utility, and the units of such quantification is called 'utils' and finally this 'utils' can be converted into money.

(3) **All other factors are constant:** It means there should be no change in income, tastes, preference, habit, fashion etc.

(4) **No time gap or no interval between consumption:**
**Ex.** MU will not reduce if 1st cup tea consume at morning and second cup tea consume at evening.

(5) **Goods should be identical/homogeneous:**
**Ex.** If 2nd Cup of tea is good quantity against 1st cup tea then MU may increase in 2nd cup.

(6) **Standard unit:**
**Ex.** Tea should be consume in standard unit of 'cup' not consumed with spoon.

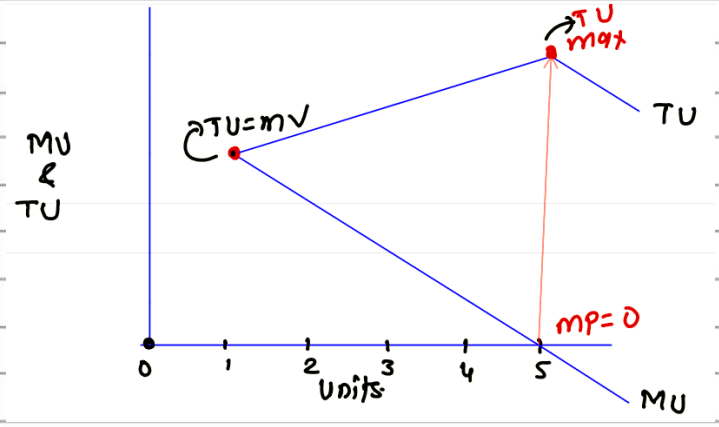
(7) **Marginal utility (MU) of money is constant:** Throughout its consumption.

(8) **Hypothesis (concept) of independent utility:** it means ignore complementary between goods.

## 2.3.2 THE LAW OF DIMINISHING MARGINAL UTILITY (MARSHALL)

"As a consumer increase consumption/ stock of any goods the marginal utility of it will diminish".

| Tea cups | MU | TU |
|---|---|---|
| 1 | 50 | 50 |
| 2 | 40 | 90 |
| 3 | 20 | 110 |
| 4 | 10 | 120 |
| 5 | 0 | 120 |
| 6 | -10 | 110 |




**Conclusion from above table and graph:**

(1) TU = MU at first unit of consumption.

(2) TU will be maximum when MU = 0 (satiation point).

or

TU curve will be at its highest when MU curve cut 'x' exis.

(3) MU can be positive / zero/ negative but TU can only be positive.

(4) After first unit of consumption "MU" start diminishing but TU start increasing till MU = 0.

(5) MU is the slope of 'TU' or we can say MU is the rate of change of TU.

## 2.3.3 LIMITATIONS / EXCEPTIONS OF LAW OF DIMINISHING MU:

**(1) Unrealistic assumptions:**

Ex. (a) cardinal measurability

(b) constant MU of money

(c) concept of independent utility.

(2) **This law may not apply to some situations:**

(a) Prestigious goods: Ex. Gold / cash etc.

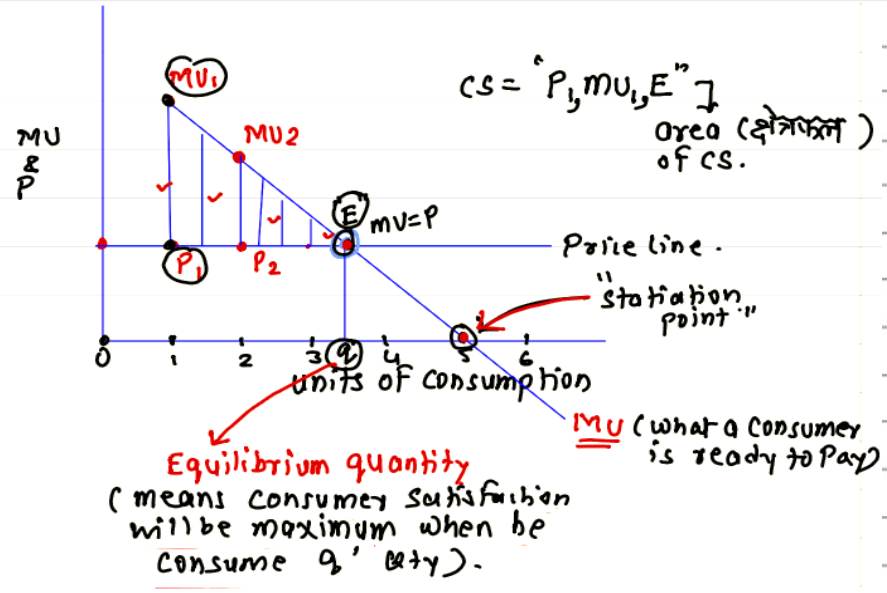
(b) Hobbies, rare collections, seeking knowledge.

(c) Music, painting etc.

(d) Habit: Ex. Alcohol, Cigarettes etc.

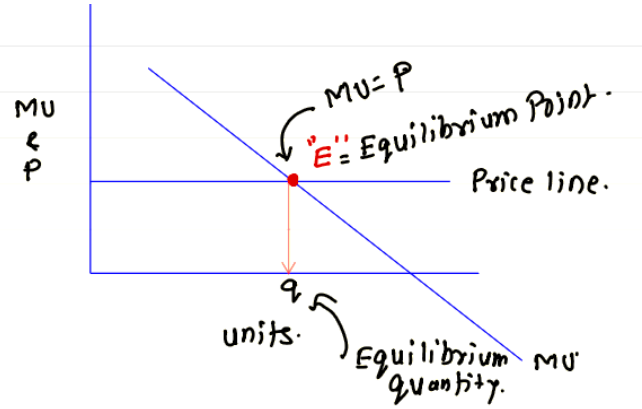
## 2.3.4 CONSUMER SURPLUS:

What a consumer is ready to pay [i.e. MU] *less* what he actually pay [i.e. price], is consumer surplus. ∴ CS = MU – P




**Consumer equilibrium in single commodity:**

(a) In single commodity consumer will be in equilibrium (i.e. maximum satisfaction) if he purchase a quantity where "MU = P".




MU=P at q units

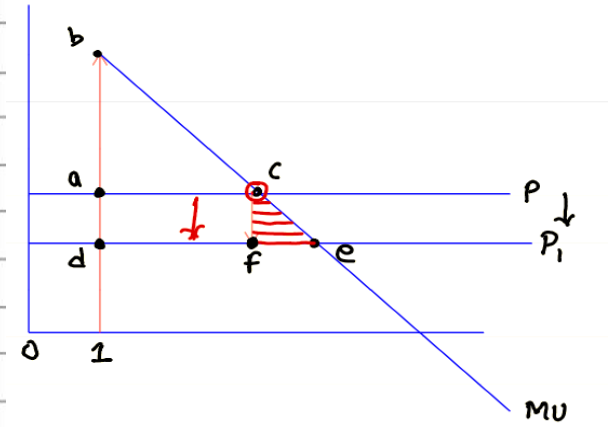
∴ Marginal utility of money ($MU_m$) → $MU_m = \frac{MU}{p}$=1.

In simple wording a consumer is obtaining MU of Rs. 1 by spending Rs. 1 at $q^{th}$ quantity.

**(b)** Consumer Equilibrium in more then one commodity "For Example consumer is purchasing goods x and y then he will be in equilibrium when MU of both goods are equal" **i.e. $MU_x = MU_y$.** This concept is explained by Marshall in "Equi-marginal utility" concept.

$$MU_m = \frac{MUx}{Px} = \frac{MUy}{Py} = 1 \text{ or } MU_m = \frac{MUx}{MUy} = \frac{Px}{Py}$$

**Effect on consumer surplus due to fall in the price**




**Conclusion:**

(a) At price P, the CS is a,b,c (area)

(b) At price $P_1$, the CS is d,b,e (area)

(c) Extra CS due to reduction in price from P to $P_1$ will be: d,a,c,e. d,a,c,f (rectangle) is CS for existing buyer and f,c,e (triangle) CS of new buyer.

## 2.4.1 APPLICATIONS OF THE CONCEPT OF CONSUMER SURPLUS

(1) Measure the welfare of consumer: The consumer having large 'CS' is likely to repeat the demand.

(2) To make the better decision about setting price and price discrimination.

(3) For large scale investment decision.

(4) For rising product price.

(5) Guide to the government to increase taxation (ex. GST).

## 2.4.2 LIMITATION OF CONSUMER SURPLUS

(1) Difficult to measure the 'MU' in terms of money.

(2) In case of necessaries or very important goods or habituated goods, the consumer surplus will be infinite. (MU – P = CS)

(3) 'CS' will be change on availability of substitute.

(4) MU of money may not be constant.

(5) CS can not be calculated for prestige goods.

## 2.5 INDIFFERENCE CURVE ANALYSIS

[J.R. Hicks & R.G.D. Allen]

**Indifference schedule:**

| Combination | Food | Cloth | MRS |
|---|---|---|---|
| A | 1 | 12 | - |
| B | 2 | 6 | 6 |
| C | 3 | 4 | 2 |
| D | 4 | 3 | 1 |

**Notes:**

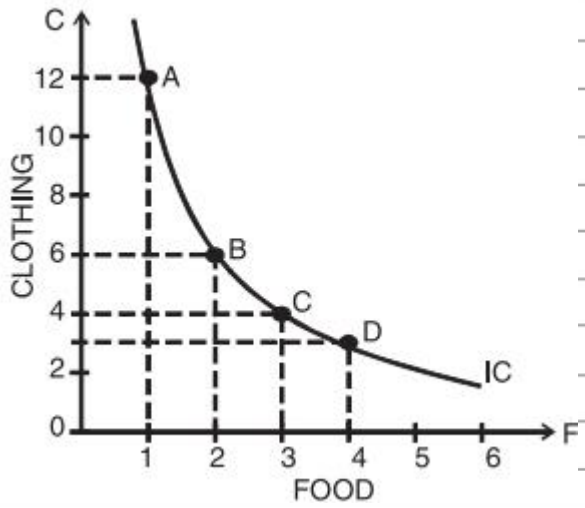
(1) MRS = Marginal rate of substitution.

(2) When a consumer is moving form combination A to D then he is sacrificing cloth for food of one unit ∴ this is called "MRS".

(3) MRS is reducing shows that the satisfaction of food is reducing on increasing its quantities.

(4) The curve created from above schedule will be called indifference curve (IC).

**Indifference curve (IC)**




**Concept of 'IC'**

(1) 'IC' curve is always negative slope, because we have to reduce cloth for extra unit of food, to maintain same satisfaction.

(2) 'IC' curve is the combination of two goods which gives same satisfaction.

(3) The shape of 'IC' curve is normally convex because MRS is reducing.

**Different shape of 'IC' curve**

MRS↓

Shape – convex
Slope – negative
(normal nature & shape of 'IC' curve)

MRS↑

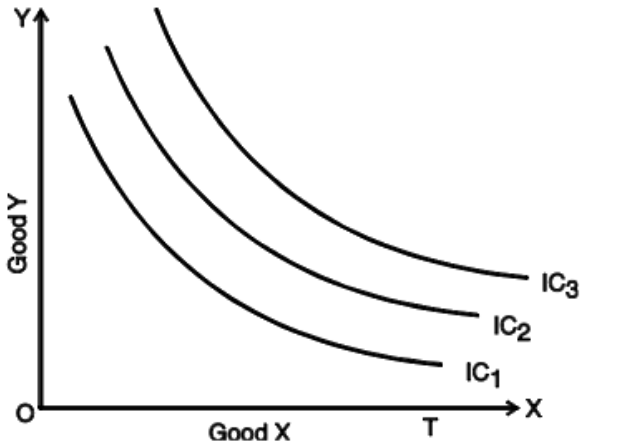
Shape – concave
Slope – negative

MRS =

Shape =Straight line
Slope – negative
(when good x & y are perfect substitute)

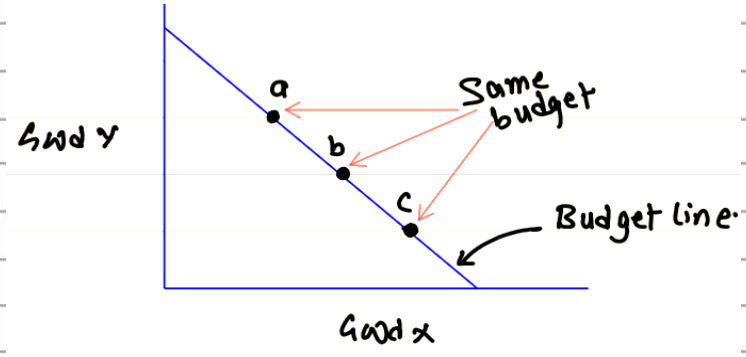
## 2.5.2 INDIFFERENCE CURVE MAP (IC MAP)

"A set of 'IC' curve is called IC map" and higher 'IC' will give more satisfaction.



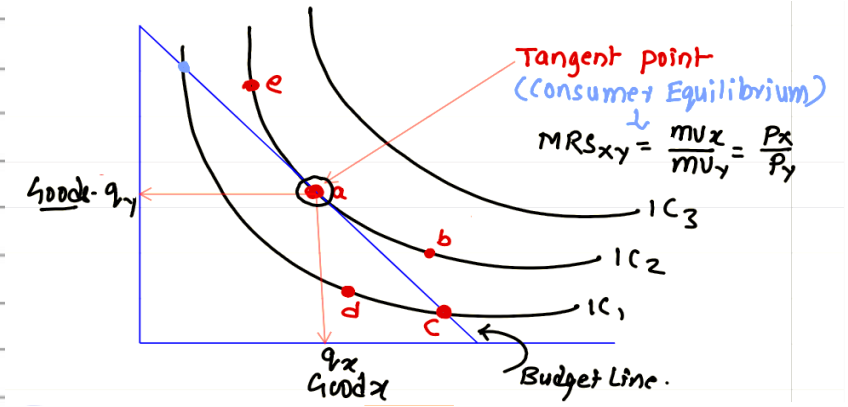

## 2.5.5 BUDGET LINE / PRICE LINE:

Each and every point of budget line presents the combination of two goods have same budget.




## 2.5.6 CONSUMER EQUILIBRIUM

**(In simple wording consumer is in maximum satisfaction)**




**Conclusion:**

(1) Consumer will get maximum satisfaction at the point where budget line touches to 'IC' curve.

(2) Therefore consumer will consume $q_x$ quantity of good x and $q_y$ quantity of good y to gain maximum satisfaction with given budget.

(3) At consumer equilibrium, we can not rearrange the quantity of good x & y.

## 2.5.0 ASSUMPTIONS OF 'IC' APPROACH (H&A)

(1) Consumer have full information and he can give preference order to goods.

(2) Consumer is rational: he prefer more quantity so he prefer higher 'IC'.

(3) Utility is only ordinally expressible (as per Marshall, utility is cardinal concept)

(4) Consumer choice are transitive:

**Ex.** If 'A' is better then 'B' and 'B' is better then 'C', then always A will be better then 'C' and therefore we can create a preference order like A, B, C.

## 2.5.3 MARGINAL RATE OF SUBSTITUTION [MRS]

(1) 'MRS' is the rate of exchange of good 'Y' for good 'X'.

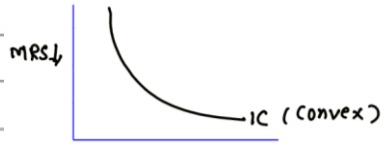
(2) Nature of 'MRS'

(i) **MRS ↓** : normally MRS is reducing, therefore 'IC' curve will be convex and the reason is:

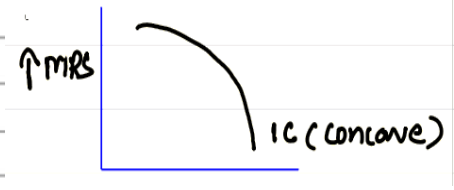
(a) want of 'x' goods is satiable and increase of good 'x' will reduce the intensity of want, therefore we prepare to give less units of good 'y'.

(b) good x & y are not perfect substitute.

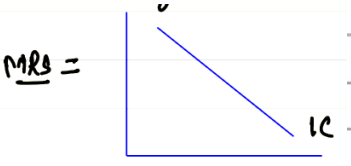
(c)



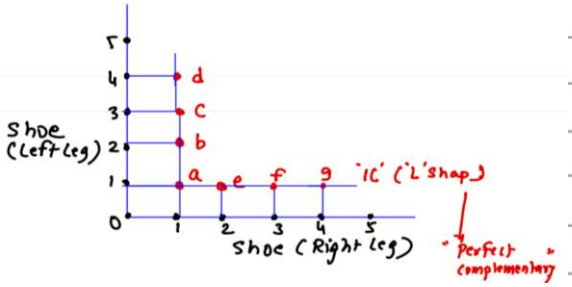

(ii) **MRS ↑** : It is possible only when utility increases when quantity increases. Ex. Habit, prestige goods, music etc. In this case IC curve will be concave.




(iii) **MRS is constant:** If goods 'x' & 'y' are perfect substitute. (IC curve will be straight line).




(iv) **'L' shape IC curve:** This is possible if goods x & y are perfect complementary.



**Reasons for MRS ↓ :**

1. The want for a particular good is satiable so that when a consumer has more of it, his intensity of want for it decreases. Thus, in our example, when the consumer has more units of food, his intensity of desire for additional units of food decreases.
2. Most goods are imperfect substitutes of one another. MRS would remain constant if the could substitute one another perfectly.

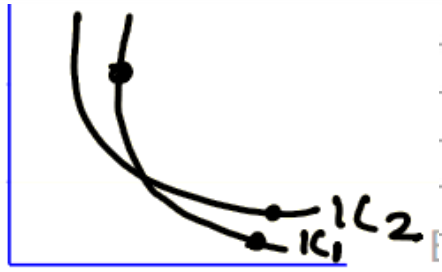
## 2.5.4 PROPERTIES (NATURE) OF INDIFFERENCE CURVE

(1) 'IC' slope downward to right (or say negative slope) we have to reduce goods 'y' to gain good x to maintain same level of satisfaction.

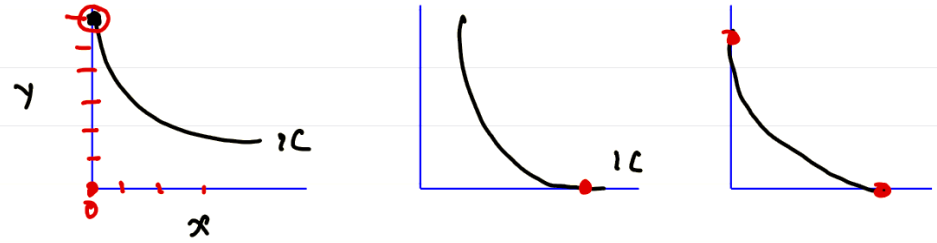
(2) 'IC' are always (normally) convex to the origin.

(3) IC may be concave, straight line & 'L' shaped.

(4) 'IC' can never intersect (cut) each other because higher IC will give higher satisfaction.




(5) 'IC' will not touch 'x' axis / 'y' axis because it result is any one goods become 'zero', but as per 'IC' approach we have to prepare combination of two goods.



# CHAPTER – 2 : THEORY OF DEMAND & SUPPLY

## UNIT III: SUPPLY

### 3.0 INTRODUCTION

Supply refers to amount of goods or service that a supplier is *willing and able to offer* to the market at various prices during given period of time.

NOTES-

1) Supply may not be equal to actual sale, same as demand may not be equal to actual purchase.

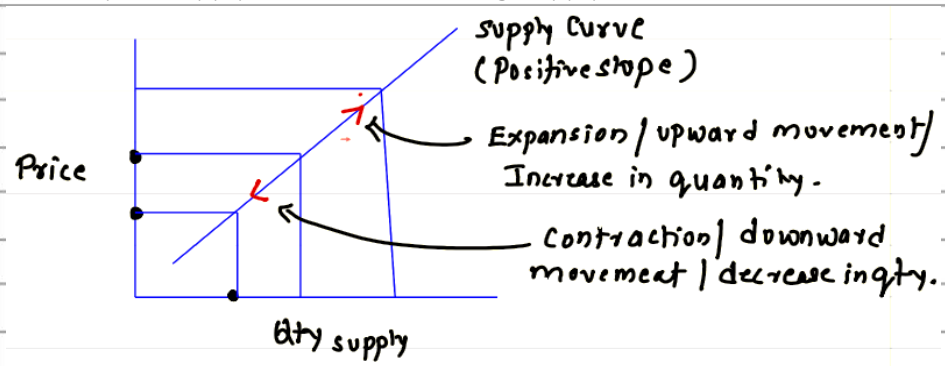
2) Demand and supply both are flow concept because we check it for different price at given period of time.

3) Willingness of supply depends on profit and ableness of supply depends on stock/ manufacturing capacity etc.

### 3.1 DETERMINANTS OF SUPPLY

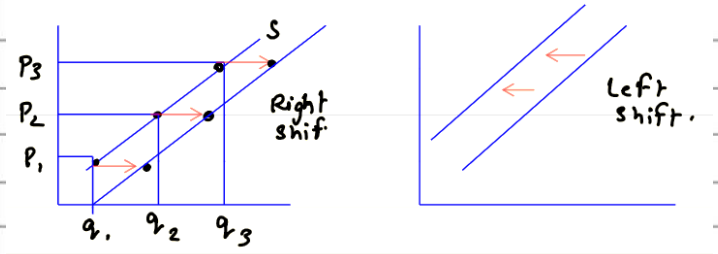
(factors responsible for change in supply)

(a) If change in supply is due to price then it will be expansion / contraction of supply or we can say increase/ decrease in quantity of supply or movement along supply curve.




∴ if price increase the quantity of supply will also increase and visa versa and if other factors are constant then it will be "law of supply."

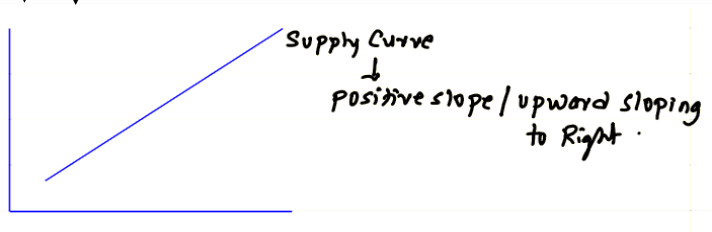
(b) If change in supply due to other factors then it will be increase / decrease in supply and there is shifting of supply curve.



i. **Price of goods :**

P ↑ S ↑ (Expansion)

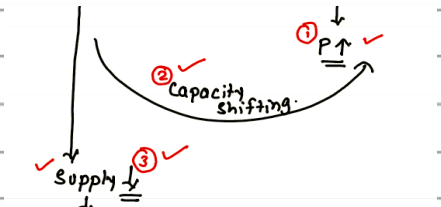
P ↓ S ↓ (Contraction)




ii. **Price of related goods :**

**Example:** Related goods (substitute) for supplier

| | | | |
|---|---|---|---|
| a) | Wheat | vs. | rice. |
| b) | Comic book | vs. | novels. |
| c) | Rice | vs. | corn |
| d) | Plastic chair | vs. | plastic toys. |




Decrease in supply due to increase in price of substitute goods.

iii. **Price of "factors of production"**

Factors of production (land , labour, capital etc.) create cost and if price of it increase the "cost of product " will also increase therefore our profit will decrease. Hence supplier will reduce supply.

**Conclusion:** price of factors of production ↑ → ↓ supply.

iv. **State of technology:** due to improvement in technology / inventions / innovation the supply will increase due increase in ability of supply.

v. **Govt. policy :**

| | | | |
|---|---|---|---|
| a) | Taxation ↑ | supply ↓ | |
| b) | Subsidy ↑ | supply ↑ | |
| c) | Exim policy : i) import ↑ | supply (India) ↑ | |
| | ii) export ↑ | supply (India) ↓ | |
| d) | Other policy :- RBI rule, banking, factory act. | | |

vi. **Nature of competition and size of industry :**

a) More competition ⟶ supply ↑

b) Big size industry ⟶ supply ↑

vii. **Future expectation regarding price**

a) If future expected price ↑ ⟶ current supply ↓

b) If future expected price ↓ ⟶ current supply ↑

viii. **Number of sellers :** more sellers ⟶ supply ↑

ix. **Other factors :**

a) Govt. industrial policy

b) Govt. foreign policy

c) Infrastructure facility

d) Weather

e) Labour strikes

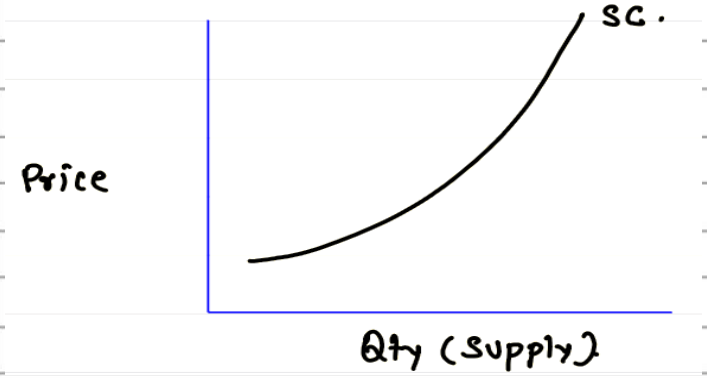
## 3.2 LAW OF SUPPLY:

If other factors are constant then P ↑ S ↑ & P ↓ S ↓

a) **Supply schedule**

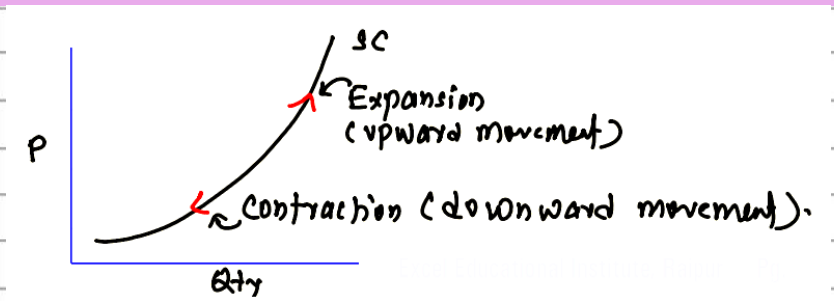
| S.NO. | PRICE (₹) | SUPPLY (QTY) |
|---|---|---|
| 1 | 10 | 100 units |
| 2 | 15 | 150 units |
| 3 | 20 | 170 units |

b) **Supply curve**

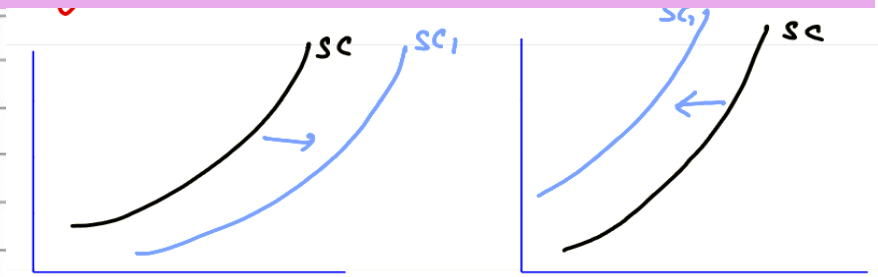



c) **Market supply schedule or market supply curve sum total of supply of all the firm of market.**

## 3.3 MOVEMENT ON SUPPLY CURVE / INCREASE OR DECREASE IN QTY SUPPLY




## 3.4 SHIFTING OF SUPPLY CURVE / INCREASE OR DECREASE OF SUPPLY




## 3.5 ELASTICITY OF SUPPLY

$Es = \frac{\%\ \Delta \text{ in quantity supply}}{\%\ \Delta \text{ in price}}$ or $Es = \frac{\Delta q}{\Delta p} \times \frac{p}{q}$

Where, p = original price

q = original quantity

**Example:** Price old ₹ 2,000 New ₹ 2,100

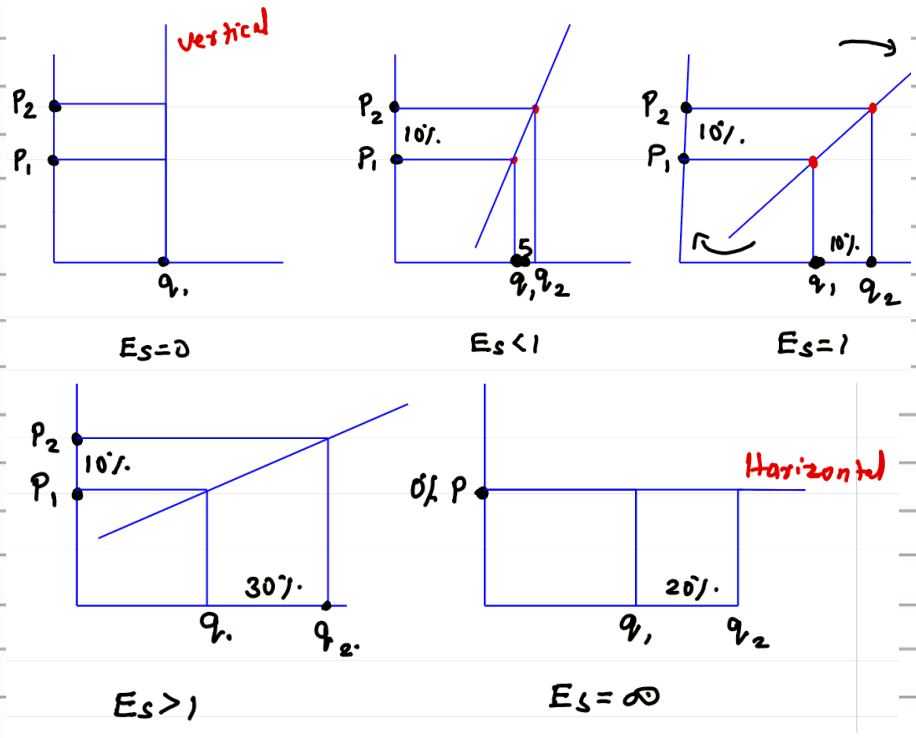
Supply old 2,500 unit New 3,000 units

**Solution:**

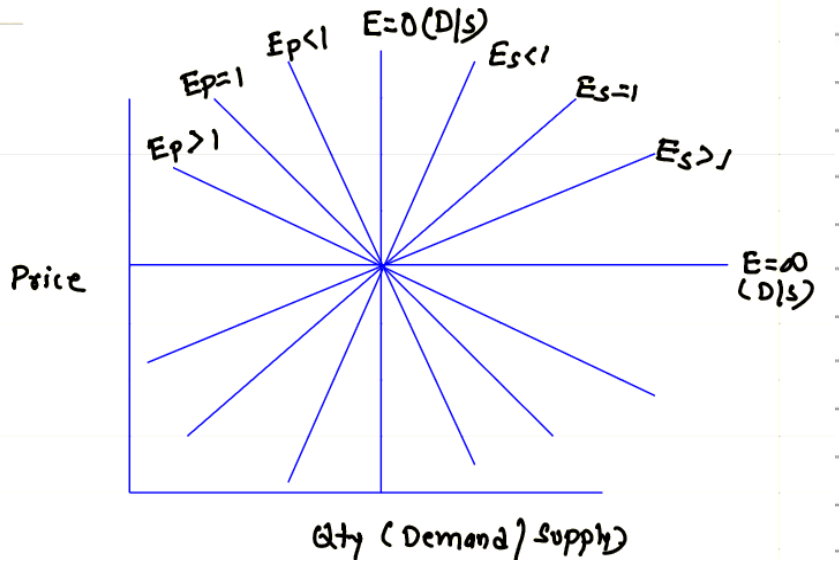
$Es = \frac{\Delta q}{\Delta p} \times \frac{p}{q} = \frac{500}{100} \times \frac{2000}{2500} = 4$

### 3.5.0 Types of supply elasticity:

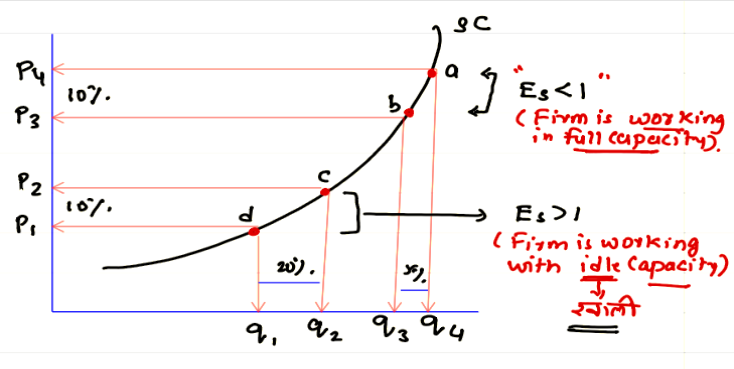
| | |
|---|---|
| Es = 0 | perfectly inelastic |
| Es < 1 | Inelastic / Relatively less elasic |
| Es = 1 | Unit elastic |
| Es > 1 | Elastic / Relatively more elastic |
| Es = ∞ | Perfectly elastic |




**Concept:** When elasticity of supply increase from Es = 0 to Es = ∞, then vertical supply curve will become horizontal clock wise.



**Different Elasticity in same supply curve**




**Conclusion:** If Firm is working in full capacity then it is very hard to increase production with respect to price increase so this is a situation of Es < 1 and if firm is working with idle capacity (ex. Capacity 500 units but present production is 300 units) then if price increases the firm can increase production more then price increase so its Es > 1.

## 3.5.1 MEASUREMENT OF ELASTICITY OF SUPPLY

(a) **Point Elasticity:** it is calculated on a single point on supply curve.

$Es = \frac{dq}{dp} \times \frac{p}{q}$ Where, p = original price

q = original quantity

"dq/dp" is differentiation of supply function with respect to price.

**Ex.** If the supply function is "q = - 100 + 10p", find the elasticity of supply, if price is Rs. 15/-.

**Sol.** P = 15, q = -100+ (10 x 15) = 50 dq/dp = 10

$Es = \frac{dq}{dp} \times \frac{p}{q} = 10 \times \frac{15}{50} = 3$ elasticity of supply

(b) **Arc Elasticity:** (Mid point method) this elasticity is also calculated between two point of supply curve but instead of taking original P&Q we will take average of original & new P & Q.

$Es = \frac{Q_2 - Q_1}{Q_2 + Q_1} \times \frac{P_2 + P_1}{P_2 - P_1}$

**Ex.** Compute arc elasticity in following case-

| | Old | New |
|---|---|---|
| Price | 12 | 15 |
| Quantity | 20 units | 50 units |

**Ans. 3.857**

## 3.5.2 DETERMINANTS OF ELASTICITY OF SUPPLY (Factors / Reasons of high / low Es)

(1) (a) If we want to increase production from current level and after this level the cost will increase very high (i.e. substantial increase in cost) ∴ Es < 1.

(b) **Complex production process:** (Ex. Production of Aircraft/ ship) ∴ Es < 1.

(c) **Production required larger time:** Es < 1.

(2) (a) **Large number of producer:** Es > 1.

(b) **High degree of competition:** Es > 1.

(c) **Entry barriers for new firm:** Es < 1.

(3) **Spare / idle production capacity:** Es > 1.

(4) **Key raw material / Input etc. are easily available:** Es >1.

(5) **Stock can be easily and inexpensively stored:** Es > 1.

(6) **Factor substitution:** supplier can easily shift labour / machine etc. from one product to another ∴ Es > 1.

(7) **If Production required very skilled worker:** Es < 1.

(8) **Continuously produce product:** Es > 1.

(9) **If future price expectation to be high then current price increase:** Es < 1.

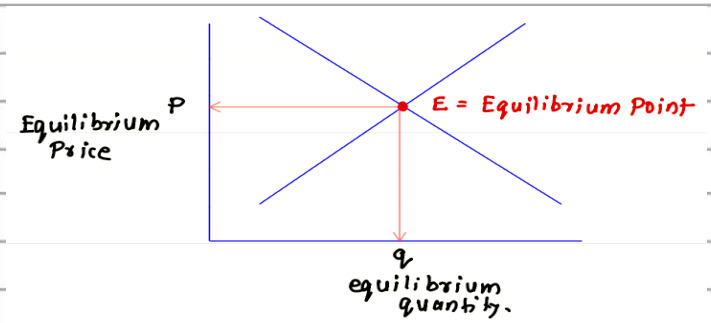
## 3.6 EQUILIBRIUM PRICE (EQUILIBRIUM MARKET PRICE OR MARKET CLEARING PRICE)

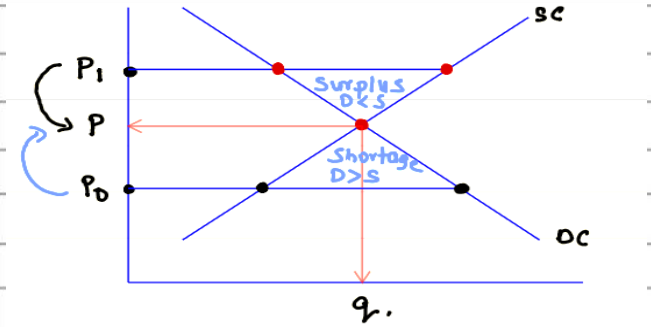
"A price in which quantity of demand is equal to quantity of supply"

**Demand and supply schedule**

| | Price | Demand | Supply | DVs | Impact on price |
|---|---|---|---|---|---|
| A | 5 | 6 | 31 | D < S | P ↓ |
| B | 4 | 12 | 25 | D < S | P ↓ |
| C | 3 | 19 | 19 | D = S | Equilibrium |
| D | 2 | 25 | 12 | D > S | P ↑ |
| E | 1 | 31 | 6 | D > S | P ↑ |

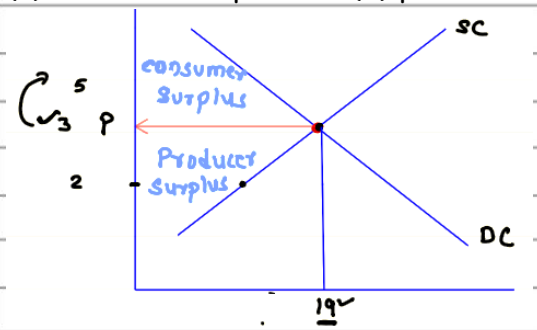
**Conclusion:** Equilibrium price in the above schedule will be Rs. 3 and 19 units will be called "Equilibrium quantity".




## 3.6.1 MARKET EQUILIBRIUM & SOCIAL EFFICIENCY

Social efficiency represents net gain to the society from all exchanges (sell/ purchase) in a market social efficiency contains (a) consumer surplus and (b) producer's surplus.




(a) **Consumer surplus:** This is the area below demand curve but above equilibrium price. Consumer surplus is difference between what a consumer is ready to pay and what he actually pay.

(b) **Producer's surplus:** This is the area above supply curve but below equilibrium price. In other words it is actual selling price less price at which he is ready to sell.

**Conclusion:** When the market is in equilibrium, the Social efficiency is achieved for both producer and consumer, therefore both party enjoy maximum possible surplus.

# CHAPTER – 3 : THEORY OF PRODUCTION AND COST

## UNIT I: THEORY OF PRODUCTION

### 1.0 MEANING OF PRODUCTION

"Production is economic activity which creates or add utility."

**According to James Bates and J.R. Parkinson** "Production is the organized activity of transforming resources into finished products in the form of goods and services; and the objective of production is to satisfy the demand of such transformed resources".

**Various process of production by:**

(i) **Changing the form:** of natural resources: Ex. Raw material to finished goods.

(ii) **Changing the place:** Ex. Shifting of sand from river site to construction site.

(iii) **Changing the time:** Ex. Cold storage making available a product when it is normally not available.

(iv) **Personal skill:** Ex. CA, Doctor, Teachers.

### 1.1 FACTORS OF PRODUCTION [INPUTS]

(a) Land (Rent) (b) Labour (Wages) (c) Capital (Interest) (d) Entrepreneur (Profile /loss).

### 1.1.0 LAND

It does not only mean land surface but includes "all free gift of nature" like air, sun lite, water etc.

**Nature:**

(i) Free gift of nature

(ii) Supply of land is fix: perfectly inelastic for economy but elastic for firm. (Economy can not change the supply of land but a firm can change it)

(iii) Indestructible power:

(iv) Land is a passive factor but labour is active factor.

(v) Land is immobile: we can not shift land form one place to another.

(vi) Land has multiple uses

(vii) Land is heterogeneous

## 1.1.1 LABOUR

'All human effort for economic reward are labour'

**Ex.** Maid servant are treated as labour but services of house wife are not labour because not for economic reward.

**Nature:**

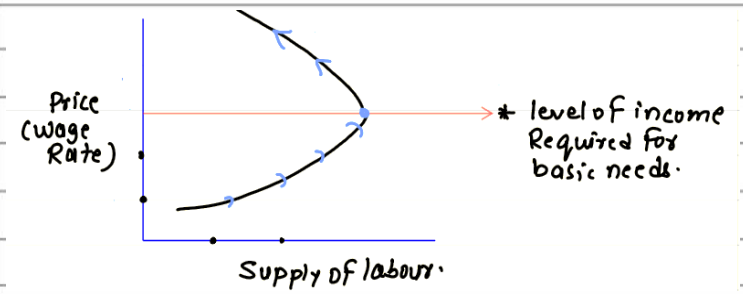
(1) Human effort

(2) Active factor

(3) Labour is perishable: Work is perishable and it can not be sold after end of the days.

(4) Poor bargaining power: due to point (3) above.

(5) Choice between "hours of labour and hours of leisure"




The above shape of supply curve of labour is call "backward bending shape"

## 1.1.2 CAPITAL

"Capital is a wealth used for further production of wealth"

**Ex.** Cash hold in home is only wealth but not capital but deposited in bank for interest it become capital.

**Stages of capital formation:**

(1) **Saving:** Saving can be effected by income or government policies like taxation.

(2) **Mobilisation of saving:** Conversion of saving into investment through financial institution / bank etc.

(3) **Investment:** investment by businessman.

**Types of capital**

(1) **Fixed capital:** Ex. Plant & machine / Building etc.

(2) **Circulating capital:** Ex. Working capital i.e. raw material.

(3) **Real capital / Tangible capital:** Ex. Building

(4) **Intangible capital:** Ex. Goodwill

(5) **Individual capital:** Ex. Home

(6) **Social capital:** Road, Bridge, Dam etc.

(7) **Human capital:** Ex. Human skill & abilities.

## 1.1.3 ENTREPRENEUR

**Function (work)**

(1) Initiating business and resource co-ordination

(2) Risk bearing: 'Risk of loss'

(3) Innovation: most important function

**Enterprise's objective and constraints:**

(1) **Organic objective:** 'To survive or stay alive'

In simple wording the revenue of the business should be enable to recover its cost.

(2) **Economic objective:** 'profit maximisation'

(3) **Social objective**

(i) Continuous supply of good product

(ii) Avoid profiteering (i.e. very high profit) and avoid antisocial activity.

(iii) Create gainful employment.

(iv) Control pollution.

(4) **Human objective**

(i) Give fair deal to employee and develop new skill

(ii) Give opportunity to employee to participate in decision making

(iii) Making job interesting and challenging.

(5) **National objective:**

(i) Employment generation

(ii) Produce as per national priorities

(iii) Skill development etc.

(iv) Becoming self dependent

**Problems / constraints in achieving above objective:**

(1) **Lack of knowledge and information:** In the changing circumstances, it is very difficult to determine 'profit maximizing price.'

(2) **Restriction imposed by government in public interest**

(3) **Infrastructure inadequacies**

(4) **Change in business and economic conditions**

(5) **Inflation rising interest rate, unfavorable exchange rate fluctuation etc.**

(6) **Conflict between objectives**

(7) **Problem relating to location and size of the plant**

(8) **Problem relating to finance**

(9) **Problem relating to organization structure**

(10) **Problem relating to marketing:** decision regarding '4P's' i.e. Product/ Promotion /Price/ Place.

(11) **Problem relating to legal formalities**

(12) **Problem relating to industrial relation**

(13) **Problem relating to selecting & organizing physical facilities**

## 1.2 PRODUCTION FUNCTION

(1) It is a technological relationship between input & output.

(2) Output is dependent variable and input is independent variable i.e. Q = f (a, b, c.....n) [Q = Production, a,b,c etc. inputs]

(3) Assumptions of production function

(4) For analysis purpose only labour and capital as a input will be considered Q = f (L, C)

**Long Run:** Long run is minimum time required to change all inputs (factors of production). In other words a minimum time in which we can change the capacity of a factory. In long run all inputs are variable.

**Short Run:** Short run is a time shorter/ less then long run. In short run at least one factor must be fix, therefore in shot run we have variable & fix factor both.

## 1.2.0 Law of variable proportion (or say law of diminishing return)

(A) **Production schedule**

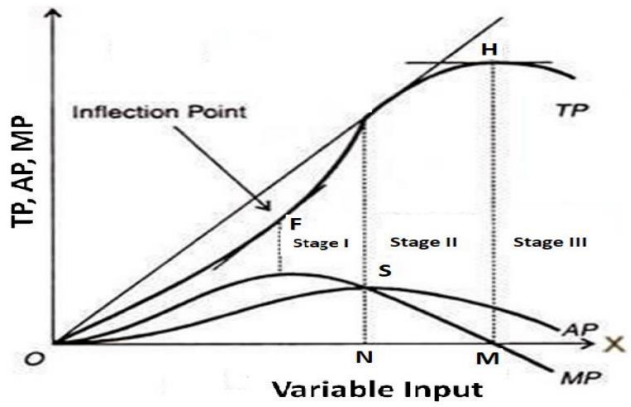
| (1) Variable input (labour) | (2) Total production (TP) | (3) Average production (AP) | (4) Marginal production (MP) |
|---|---|---|---|
| 1 | 100 | 100 | 100 |
| 2 | 210 | 105 | 110 |
| 3 | 330 | 110 | 120 |
| 4 | 440 | 110 | 110 |
| 5 | 520 | 104 | 80 |
| 6 | 600 | 100 | 80 |
| 7 | 670 | 95.7 | 70 |
| 8 | 720 | 90 | 50 |
| 9 | 750 | 83.3 | 30 |
| 10 | 750 | 75 | 0 |
| 11 | 740 | 67.3 | -10 |

**Formula-**

**$MP_n = TP_n - TP_{n-1}$ or $MP = \Delta TP / \Delta Q$ (Input)**

**AP = TP / Input (labour)**

**Concept:** "marginal is faster than average"




(1) In this law proportion of fix input and variable input changes.

(2) Total production (TP): Total output of all inputs taken together.

(3) Average production (AP): AP = TP / No of units of variable inputs (i.e. labour)

(4) Marginal production (MP): $MP_n = TP_n - TP_{n-1}$

or

MP = Δ TP / Change in no of units of variable inputs

| | | | |
|---|---|---|---|
| | **Example:** | Labour | TP (units) |
| | | 8 | 100 |
| | | 10 | 120 |
| | | 11 | 125 |
| | | $MP_{10}$ = ?. | $MP_{11}$ = ? |

**(5) Assumptions of law of variable proportion**

(a) Technology is not changing

(b) At least one factor must be fix (i.e. short run)

(c) Proportion of fix & variable inputs are changing

(d) This law does not apply if any product require fix proportion of fix & variable inputs (ex. If 1:1 ratio required for any product of labour & capital) MP=0

(e) input & output is taken in physical units not in money term.

**Conclusion**

(1) If TP is maximum then MP is zero or we can say TP curve is highest then MP curve cut X axis.

(2) MP cut AP from its highest point (i.e. MP = AP)

(3) Before point of inflection the TP increases at high rate but after point of inflection TP increase upto point 'H' but low rate.

(4) Stage 1 end at the point where MP = AP.

(5) Stage 2 end at the point where MP = 0/TP is maximum.

(6) (i) Stage 1 = Law of increasing return

(ii) Stage 2 = Law of diminishing return

(iii) Stage 3 = Law of negative return.

## 1.2.1 Cobb- Douglas production function:

(a) This production function is given by Paul. H. Douglas and C.W. Cobb (USA). $Q = KL^{a}C^{(1-a)}$

(b) [Q = output, L = labour, C = capital, K & A is positive constant of a particular industry]

(c) Conclusion: Labour contribute $\frac{3}{4}$ & capital contribute $\frac{1}{4}$ of production.

**(B) Explanations/ Reasons of three stages**

**(1) Explanation of law of increasing return (Stage-1)**

(a) In the beginning fix factor is more then variable factor due to 'Indivisibility' of fix factor and also we start any business with high capacity, therefore by increasing

variable factor we can use fix factor more intensively and effectively.

(b) Due increasing variable factor (i.e. labour) division of labour and specialization is possible.

**(2) Explanation for law of diminishing return (Stage 2)**

Increasing variable factor after a level, the fix factor become inadequate hence MP & AP start decreasing.

**(3) Explanation for law of negative return (Stage 3)**

When variable factor is too excessive they get in each other way therefore TP start declining.

**Stage of operation**

(1) Stage 1 & 3 is called "Economic absurdity" or "Economic nonsence"

(2) A rational producer should produce in stage 2 in a point where he get maximum profit.

## 1.2.2 Law of return to 'Scale' (related to long run)

**(a) Constant return to scale**

All input ↑ 20% ⟶ output ↑ 20%

↓ 20% ⟶ ↓ 20%

**(b) Increasing return to scale**

(due to Economics to scale)

All input ↑ 20% ⟶ output ↑ 30%

↓ 20% ⟶ ↓ 30%

**(c) Decreasing return to scale** (due to diseconomies to scale)

All input ↑ 20% ⟶ output ↑ 10%

20% ⟶ ↓ 10%

**Note:**

(1) It is assumed that the ratio of all input does not change.

(2) All input are variable

(3) Positive relation between input & output in all 3 law of return to scale (but in law of variable proportion this may be positive or negative).

(4) Revised concept by Cobb-douglas for long-run: originally Cobb-douglas assumed only "Constant return to scale" but later on they rewrote the equation as: $Q = KL^a C^b$

(Q = output, L = labour, C = capital, K, a & b are positive constant as per industry)

**If**

$a + b > 1$ then increasing return to scale

$a + b = 1$ then constant return to scale

$a + b < 1$ then diminishing return to scale

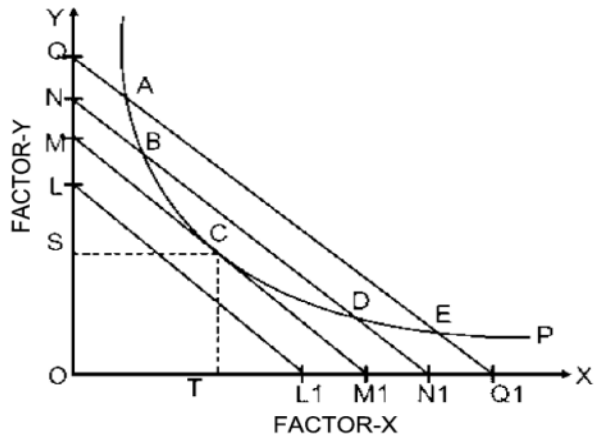
## 1.3 PRODUCTION OPTIMIZATION: [PRODUCER'S EQUILIBRIUM]

A producer can find the best combination of two inputs for a given output, so that the cost is minimum and our profit is maximum. This situation of best combination of two inputs are called producer's equilibrium.

**For above purpose we have to draw two curve:**

(a) **Equal production curve (may be called Isoquant/ Production indifference curve/ Iso-product curve):** each and every point on isoquant is the combination of two input which give same output.

(b) **Equal cost curve (may be called Iso-cost):** each and every point on iso-cost is the combination of two input which give same cost.




(1) MRTS = Marginal rate of technical substitution

(2) Isoquant is convex because MRTS is reducing

(3) Why MRTS reduces?, because input X input Y are not perfect substitute.

(4) Producer will be in equilibrium when isoquant touches isocost.

(5) When we move from one point to another in isoquant curve then (a) ratio of both input change (b) marginal production of each input change (c) MRTS also change.

# CHAPTER – 3 : THEORY OF PRODUCTION AND COST

## UNIT II: THEORY OF COST

An amount required to produce any product is the cost in simple wording.

## 2.0 COST CONCEPT (TYPES OF COST)

### (a) Accounting cost and economic cost

**Accounting cost:** A cost for which we make entry in books of accounts. Ex. Wages, rent, interest, depreciation.

**Economic cost:** Accounting cost + opportunity cost (opportunity cost is the cost of missed opportunity or profit forgone of missed opportunity). Ex. Interest of own fund, rent of own building.

### (b) Outlay cost and opportunity cost

Outlay cost is a cost which we have to pay actually to other person. Ex. Salary, wages, raw material etc.

### (c) Direct / traceable cost and indirect / non traceable cost

**Direct cost** can be identified and/or traceable to a product. Ex. Raw material, Direct labour & factory cost.

**Indirect cost** can not be identified/ traceable to a product. Ex. (1) Rent of office building (2) salary of accountant (3) Advertisement expenses.

### (d) Incremental cost and sunk cost

**Incremental cost** means extra cost (marginal cost) of taking any decision.

**Sunk cost** means those cost which are already incurred, and can not be recovered.

### (e) Historical cost and replacement cost

**Historical cost:** original cost of an asset is historical cost. Ex. Value of asset presented in balance sheet at historical cost.

**Replacement cost:** cost required to replace old asset. Ex. Cost of new asset less NRV of old asset.

**(f) Private cost and social cost**

**Private cost:** Private costs are costs actually incurred or provided for by firm and are either explicit or implicit. They normally figure in business decisions as they form part of total cost and are internalized by the firm.

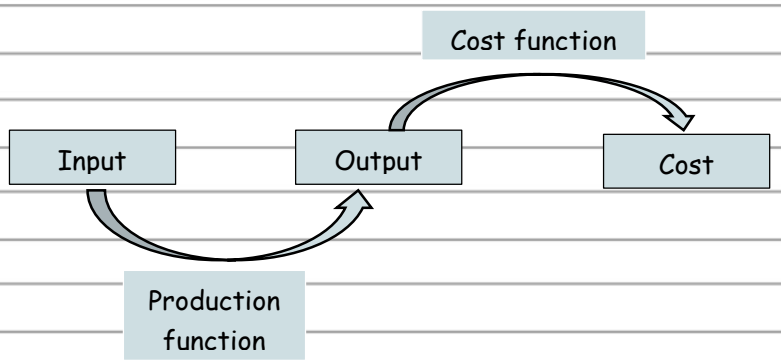
**Social cost:** social cost, on the other hand, refers to the total cost borne by the society on account of a business activity and includes private cost and external cost. It includes the cost of resources for which the firm is not required to pay price such as atmosphere, rivers, roadways etc. and the cost in terms of dis-utility created such as air, water and environment pollution.

**(g) Fixed cost and variable cost**

**Fixed cost** is not a function of output, it means when we change the quantity of production, this cost does not change. Ex. Rent of building, interest on past borrowing, salary of accountant etc.

**Variable cost** is a function of output. Ex. Raw material cost, labour cost, direct factory cost etc.

## 2.1 COST FUNCTION




Cost function is the mathematical relation between cost of a product and various determinants (mainly output).

## 2.2 SHORT-RUN TOTAL COST

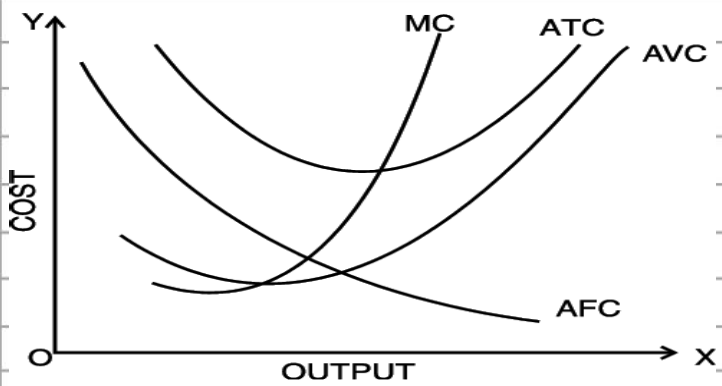
Total cost (TC) = total fixed cost (TFC) + total variable cost (TVC)

| ÷ | ÷ | ÷ |
|---|---|---|
| Output | Output | Output |
| = ATC or AC | = AFC | = AVC |

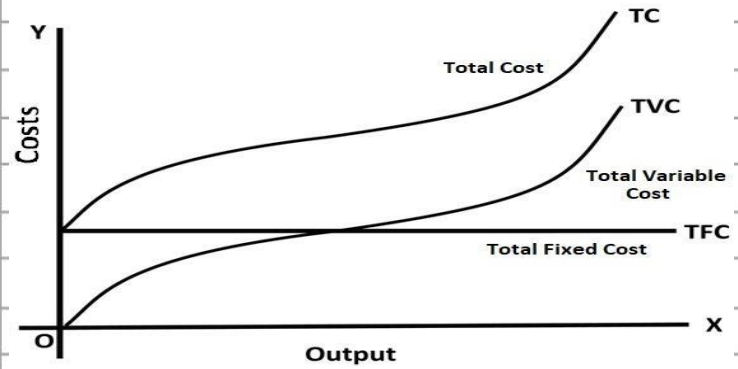
**Note:** AC = AFC +AVC

## Various Cost-

| Units of output | Total fixed cost | Total variable cost | Total cost | Average fixed cost | Average variable cost | Average total cost | Marginal cost |
|---|---|---|---|---|---|---|---|
| 0 | 1000 | 0 | 1000 | - | - | - | - |
| 1 | 1000 | 50 | 1050 | 1000.00 | 50.00 | 1050.00 | 50 |
| 2 | 1000 | 90 | 1090 | 500.00 | 45.00 | 545.00 | 40 |
| 3 | 1000 | 140 | 1140 | 333.33 | 46.67 | 380.00 | 50 |
| 4 | 1000 | 196 | 1196 | 250.00 | 49.00 | 299.00 | 56 |
| 5 | 1000 | 255 | 1255 | 200.00 | 51.00 | 251.00 | 59 |
| 6 | 1000 | 325 | 1325 | 166.67 | 54.17 | 220.83 | 70 |
| 7 | 1000 | 400 | 1400 | 142.86 | 57.14 | 200.00 | 75 |
| 8 | 1000 | 480 | 1480 | 125.00 | 60.00 | 185.00 | 80 |
| 9 | 1000 | 570 | 1570 | 111.11 | 63.33 | 174.44 | 90 |
| 10 | 1000 | 670 | 1670 | 100.00 | 67.00 | 167.00 | 100 |
| 11 | 1000 | 780 | 1780 | 90.91 | 70.91 | 161.82 | 110 |
| 12 | 1000 | 1080 | 2080 | 83.33 | 90.00 | 173.33 | 300 |




**Short Run Total Cost Curve**




**Short Run Average & Marginal Cost Curve**

## Conclusion:

(1) AVC, ATC & MC curves are 'U' shaped curve.

(2) AFC curve will never 'U' shaped but it is downward sloping to right.

(3) AVC, ATC, MC, AFC will never touches X & Y axis because value of these can not be ZERO and it can not be calculated for ZERO output.

(4) MC curve cut AVC & AC from its minimum (as marginal is faster then average).

(5) The quantity of output where ATC is minimum (i.e. MC cut ATC or MC = ATC) is called efficient level of output. (However our ultimate goal is not to minimize cost but to maximize profit.

(6) TC = TFC when output is ZERO.

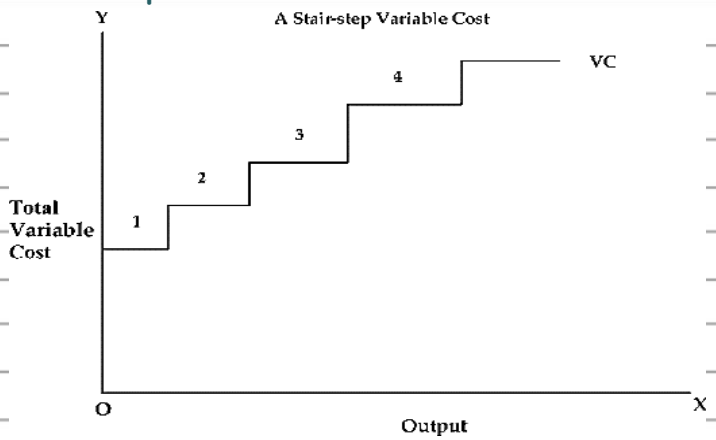
**Relationship between Average Cost and Marginal Cost:**

(1) When average cost falls as a result of an increase in output, marginal cost is less than average cost. ($AC\downarrow = MC < AC$)

(2) When average cost rises as a result of an increase in output, marginal cost is more than average cost. ($AC\uparrow = MC > AC$)

(3) When average cost is minimum, marginal cost is equal to the average cost. In other words, marginal cost curve cuts average cost curve at its minimum point **(i.e. optimum point).**

**Stair Step Variable Cost:**

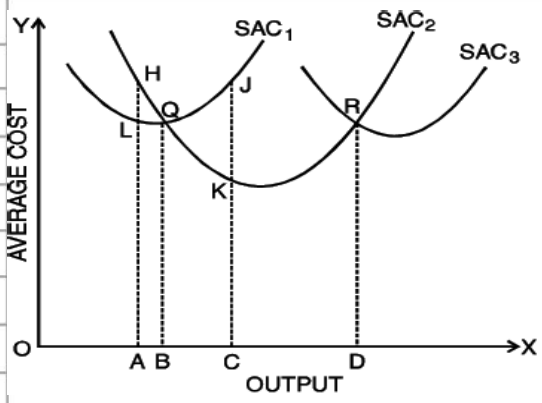



**A Stair-step Variable Cost**

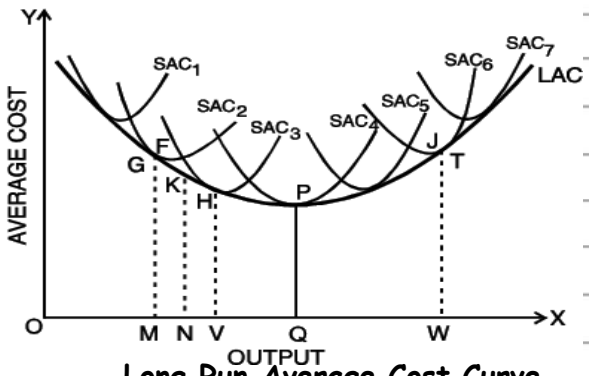
"If any cost remain fix upto a certain level of output but suddenly jump to new higher level and again fix upto that next level, then this nature of cost create stair step variable cost"

**Ex.** Cost of taking exam in class having student capacity 50.

## 2.3 LONG-RUN AVERAGE COST CURVE [LAC CURVE]




Short Run Average Cost Curve




Long Run Average Cost Curve

**Notes:**

(1) Long-run average cost curve may be called 'planning curve', 'envelope curve' & short run average cost curve may be called plant curve'

(2) LAC curve is the **tangent point** on minimum of all SAC curve.

(3) Actually LAC curve when declining it is tangent to the falling portion of SAC and, when LAC curve is rising, it is tangent to rising portion of SAC.

(4) LAC curve will be tangent at the minimum point of only one SAC at the minimum point of LAC.

(5) LAC curve is boat shape, or flatten 'U' shaped curve.

## 2.4 ECONOMIES AND DISECONOMIES OF SCALE (E/D)

Economies arising out of large-scale production can be grouped into two categories; viz., internal economies and external economies. Internal economies are those economies of production which accrue to the firm when it expands its output, so that the cost of production would come down considerably and place the firm in a better position to compete in the market effectively. Internal economies arise purely due to endogenous factors relating to efficiency of the entrepreneur or his managerial talents or the type of machinery used or the marketing strategy adopted. These economies arise within the firm and are available exclusively to the expanding firm. On the other hand, external economies are the benefits accruing to each member firm of the industry as a result of expansion of the industry.

**(A) Internal Economics & Diseconomies (E&D)**

**(i) Technical E&D**

(a) **Economies:** only in large scale we can use specialized and efficient form of machine (new technology) & labour (division of labour).

(b) **Diseconomies:** Beyond an 'optimum level' (i.e. minimum point of LAC), the long run cost will be high due to mismanagement.

**(ii) Managerial E&D**

(a) **Economies:** The concept of 'division of labour' can be apply to management also. For example large scale company can appoint different manager for different area like 1. Production manager 2. Sales manager 3. Finance manager.

(b) **Diseconomies:** after an optimum level problem of co-ordination between different manager and department and delay in decision making.

**(iii) Commercial E&D**

(a) **Economies:** We can go for bulk order for raw material, therefore the cost will be low.

(b) **Diseconomies:** After an optimum level the marketing cost to sale this huge output is more then the benefit in cost.

**(iv) Financial E&D**

(a) **Economies:** Arrangement of fund is more easy for large scale companies specially it can issue share because of goodwill.

(b) **Diseconomies:** More dependence on external finance leads to increase in the cost of raising finance.

**(v) Risk bearing E&D**

(a) **Economies:** Large company is in a better position to withstand economic ups & downs, due to multi product.

(b) **Diseconomies:** Due to multi product the company can not go highest in any product due to management issue.

**(B) External Economies & Diseconomies**

(1) **Cheaper raw material and capital equipment:** Exploration of new and cheaper source of raw material, machinery will be done only for large industry.

(2) **Technological:** discovery of new technology is only for large industry.

(3) **Skilled labour:** labour in the area of large industry are well accustomed with production of goods of that industry.

(4) **Growth of ancillary industries:** Expansion of industry encourage the growth of ancillary industry (ex. Raw material supplier, processing unit of our waste product etc.).

(5) **Better transportation and marketing facilities**

(6) **Business information:** Important information regarding product, technology, labour, price etc. is easy & fast available only in large industry.

**Diseconomies:**

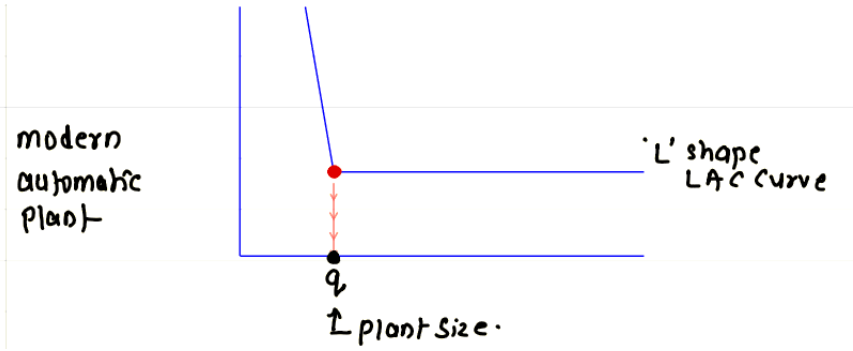
(1) After a level the government restrict the expansion of industry due to pollution etc.

(2) Requirement of various factor of production increases, therefore the price of labour, transportation, capital, land etc. is very high.

**'L' shaped long-run average cost curve**

Normally LAC curve is flattened 'U' shape or we can say "boat shape" but in modern firm using automatic machinery having 'L' shaped LAC curve.

'L' shaped LAC curve is because initially when output increases upto "Plant size" the productivity will increase but after such scale the cost per unit (AC) will be constant.



# CHAPTER – 4 : PRICE DETERMINATION IN DIFFERENT MARKETS

## UNIT I: MEANING AND TYPES OF MARKET

### 1.0 MEANING OF MARKET

While studying about market, it is essential to understand how price is determined. Since this is done in the market, we can define the market simply as all those buyers and sellers of a good or service who influence price.

As per economics following components should be available in market and it need not be a particular place:

(i) Buyer & sellers
(ii) Product or services
(iii) Bargaining for price
(iv) knowledge about market to both buyer & seller
(v) one price for a product / services at a given time

### 1.0.0 CLASSIFICATION OF MARKET

On the basis of –

**(a) Geographical area:**

(i) local market
(ii) regional market
(iii) national market
(iv) international market.

**(b) Time:**

(i) very short period market (vegetable, flowers)
(ii) long period market (normal price)
(iii) short period market (food etc.)
(iv) very long period market (secular market)

**(c) Nature of transaction:**

(i) spot/cash market
(ii) forward /future market

**(d) Regulation:**

(i) regulated market (share market)
(ii) unregulated market (local market)

(e) **Volume of business:**

(i) Wholesale market (B2B)

(ii) Retail market (B2C)

(f) **Type of competition:**

(i) Perfectly competitive market

(ii) Monopolistic competition market

(iii) Monopoly market

(iv) Oligopoly market

## 1.1 TYPES OF MARKET STRUCTURES:

| S.N. | Types of market | Seller | Buyer | Product | Price |
|---|---|---|---|---|---|
| 1 | Perfect competitive market (price taker firm) | Very large | Many | Identical | Decided by market |
| 2 | Monopolistic competition | Large | Many | Some difference | Seller |
| 3 | Monopoly | Single | Many | Unique | Seller |
| 4 | Oligopoly | Few (2-10) | Many | Any | Seller |

## 1.2 CONCEPT OF TOTAL REVENUE (TR), AVERAGE REVENUE (AR) & MARGINAL REVENUE (MR)

(a) **TR = Price (P) x Quantity Sold (Q)** (Practically TR is sales in account)

(b) **AR = TR / Q**

(c) $MR_n = TR_n - TR_{n-1}$

or

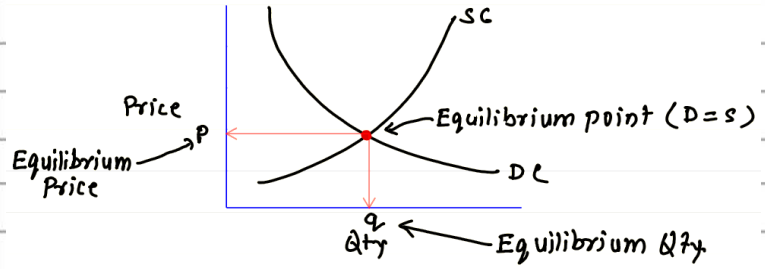
$MR = \Delta TR / \Delta Q$

**Note:** In all above cases 'Q' = Quantity sold.

**AR and MR curve in case of perfect competitive firm and other firms**

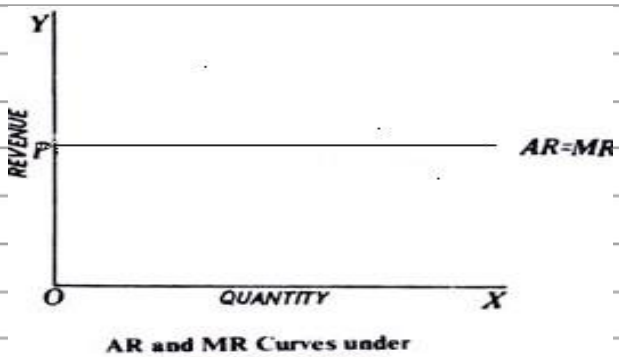
(a) **Perfect competitive firm (PC firm)[Price Taker]:** Perfect competitive firm have to adopt price of market, therefore it is called price taker firm.

Price in the market will be decided on the basis of demand & supply as follows:




Say the equilibrium price given by market is Rs. 5/- then all firm in the market have sale @ Rs. 5/- and their AR & MR will be as follows:-

| Quantity sold | TR | AR (TR /Q) | MR ($TR_n - TR_{n-1}$) |
|---|---|---|---|
| 1 | 5 | 5 | 5 |
| 2 | 10 | 5 | 5 |
| 3 | 15 | 5 | 5 |
| 4 | 20 | 5 | 5 |
| 5 | 25 | 5 | 5 |




AR and MR Curves under Perfect Competition

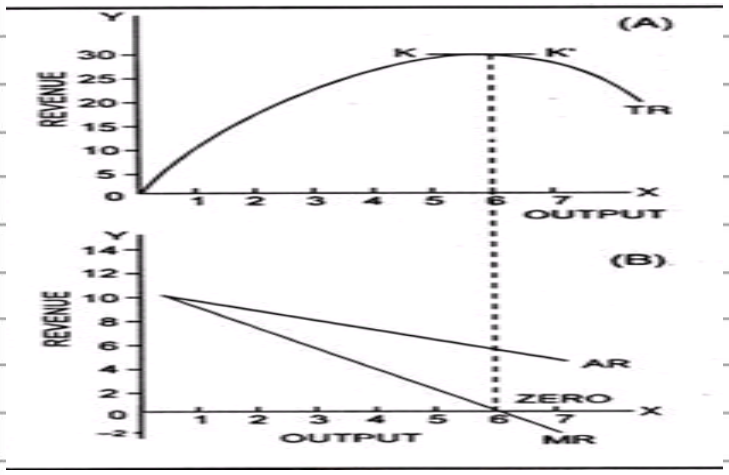
**Conclusion**

The AR & MR curve in PC firm is parallel to x axis and also identical.

**AR & MR curve for MC firm and monopoly**

These firms are not price taker so they have to reduce price if they want to sell more (law of demand)

| Quantity sold | Price (AR) | TR | MR |
|---|---|---|---|
| 1 | 10 | 10 | 10 |
| 2 | 9 | 18 | 8 |
| 3 | 8 | 24 | 6 |
| 4 | 7 | 28 | 4 |
| 5 | 6 | 30 | 2 |
| 6 | 5 | 30 | 0 |
| 7 | 4 | 28 | -2 |




**Total Revenue, Average Revenue and Marginal Revenue Curves of a Firm which has downward Sloping Demand Curve**

AR & MR both curve downward sloping curve and MR curve is below AR curve and in case of monopoly MR curve is just half way between AR curve & Y axis.

## 1.2.1 RELATIONSHIP BETWEEN AR, MR, TR & PRICE ELASTICITY OF DEMAND

Marginal revenue, average revenue and price elasticity of demand are uniquely related to one another through the formula:

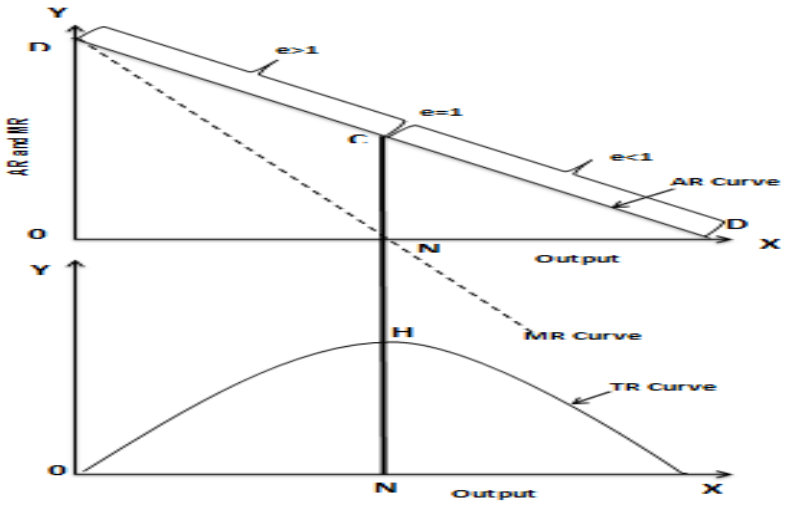
$$MR = AR \times \frac{e-1}{e}$$

Where e = price elasticity of demand

Thus if e = 1, MR will be ZERO

and if e > 1, MR will be positive

and if e < 1, MR will be negative




**Relationship between AR, MR, TR and Price Elasticity of Demand**

## 1.2.1 BEHAVIOURAL PRINCIPLES

(a) **Principle 1:** A firm should stop production if TR < TVC or AR < AVC or a firm should produce if TR ≥ TVC or AR ≥ AVC

**Example 1:** output 10,000 unit, sale price (AR) = Rs. 8/-, Purchase price (AVC) = Rs. 10/-, TFC = Rs. 30,000 /-

**(a) If we produce:**

| | |
|---|---|
| TR (8 x 10,000) | 80,000 |
| (-) TVC (10 x 10,000) | 1,00,000 |
| | (-) 20,000 |
| (-) TFC | (-) 30,000 |
| Loss | (-) 50,000 |

**(b) If we stop production**

| | |
|---|---|
| TR | Nil |
| (-) TVC | Nil |
| | Nil |
| (-) TFC | (-) 30,000 |
| Loss | (-) 30,000 |

AR (Rs. 8/-) is less then AVC (Rs. 10/-) so we should stop production to minimize loss (in short run)

**Example 2:** If in example 1 if Selling price is Rs. 11/- then

**(a) Produce**

| | |
|---|---|
| TR (11 x 10,000) | 1,10,000 |
| (-) TVC (10 x 10,000) | 1,00,000 |
| | 10,000 |
| (-) TFC | (-) 30,000 |
| Loss | (-) 20,000 |

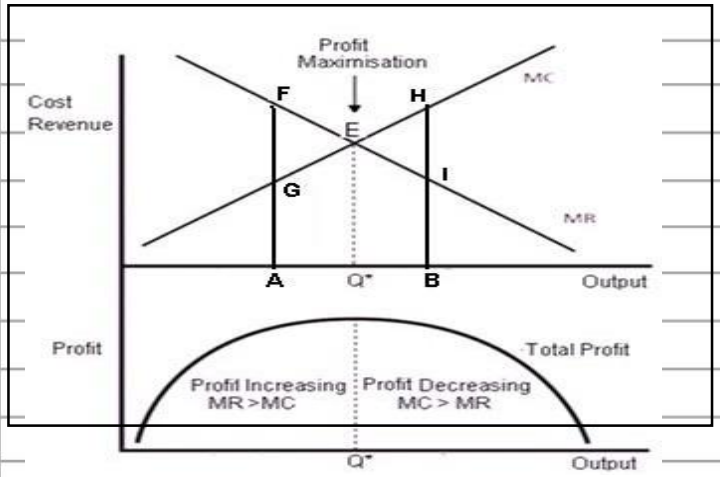
**(b) No production**

| | |
|---|---|
| TR | Nil |
| (-) TVC | Nil |
| | Nil |
| (-) TFC | - 30,000 |
| Loss | (-) 30,000 |

AR = Rs. 11/-, AVC = Rs. 10/-, AC = 13.

AR = Rs. 11 is more than AVC Rs. 10 so, we should produce

(b) **Principle 2:** A firm should produce when MR = MC (i.e. MC curve cut MR curve from below), to maximize profit (minimize loss).




**Equilibrium of the Firm: Maximization of Profits**

## Short Run Decision

| | Situation | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| | | AR<AVC | AR=AVC | AR>AVC but AR<ATC | AR=AC | AR>ATC |
| | | TR<TVC | TR=TVC | TR>TVC but TR<TC | TR=TC | TR>TC |
| | Output [units] | 10000 | 10000 | 10000 | 10000 | 10000 |
| | AR (Price) | 8 | 10 | 11 | 13 | 15 |
| | AVC | 10 | 10 | 10 | 10 | 10 |
| | | | | | | |
| 1 | TR | 80000 | 100000 | 110000 | 130000 | 150000 |
| 2 | less TVC | 100000 | 100000 | 100000 | 100000 | 100000 |
| 3 | Balance [contribution*] | (-) 20000 | 0 | 10000 | 30000 | 50000 |
| 4 | less TFC | (-)30000 | (-)30000 | (-)30000 | (-)30000 | -30000 |
| 5 | Profit or loss | -50000 | -30000 | -20000 | 0 | 20000 |

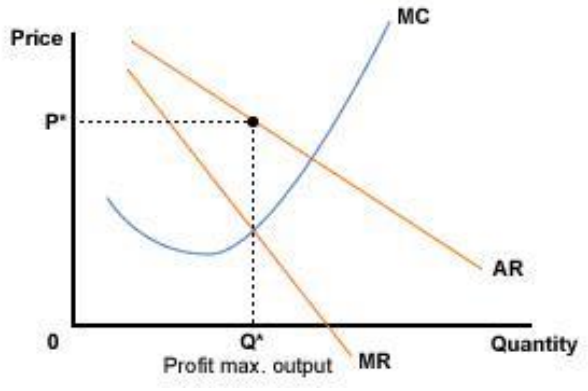
## Situation wise notes:

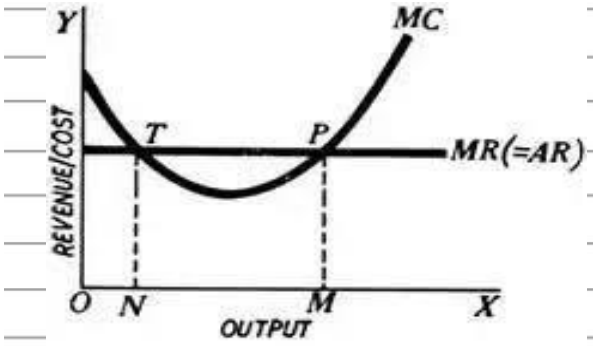
1. We should shutdown temporary as per principle 1, because if we operate our loss is 50000 if not the loss is only 30000.
2. This situation is called shutdown point because whether we produce of not our loss will be equal i.e. 30000 [here Loss = TFC].
3. We should continue even in case of loss [here ATC or AC = 130000/10000 = 13].
4. This situation may be called Break-Even Point [BEP] i.e. no profit no loss situation but in economics this situation is called normal profit because AC also includes "Normal Profit" or we can say ZERO Economic Profit.

5. This situation in economics is called "Super Normal Profit" or we can say POSITIVE Economic Profit.

**Other notes:**

1. **Equilibrium level of output (profit maximization level of output)-** If we decide to produce (i.e. not cover by principle 1) then to maximize profit we have to produce where MR = MC. (Principle 2) or we can say an output where MC curve cut MR curve from below.



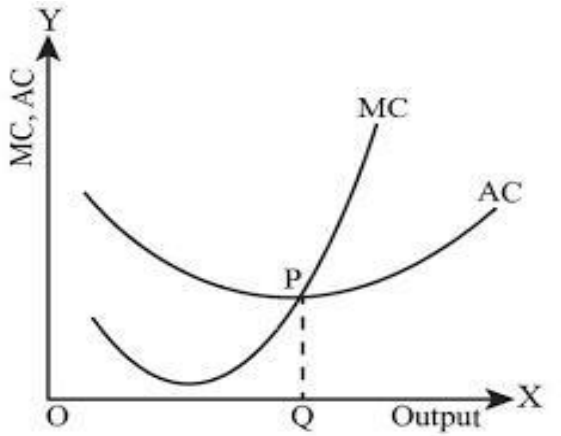




**Firm's Equilibrium Under Perfect Competition**

2. **Efficient level of output (optimal output) –** An output where average cost (AC or ATC) is minimum **or** we can say an output where MC curve cut AC curve from below.

3. AR is nothing but price and also AR curve is nothing but Demand Curve.

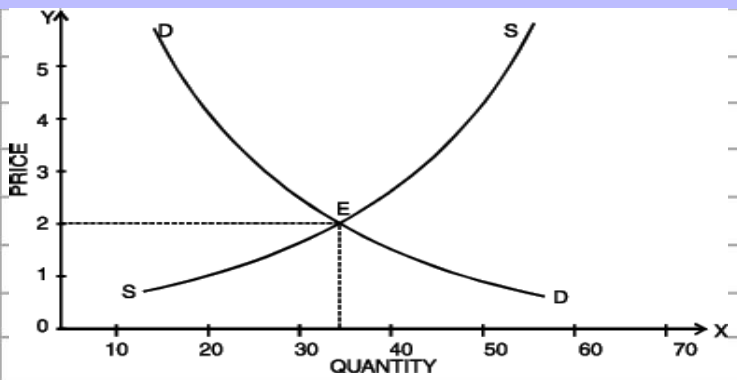
4. **Contribution** = TR - TVC, it is the contribution towards TFC.



# CHAPTER – 4 : PRICE DETERMINATION IN DIFFERENT MARKETS

## UNIT II: DETERMINATION OF PRICES

### 2.1 DETERMINATION OF PRICE- A GENERAL VIEW




**Determination of Equilibrium Price**

### 2.2 CHANGES IN DEMAND & SUPPLY

The four possible changes in demand and supply are:

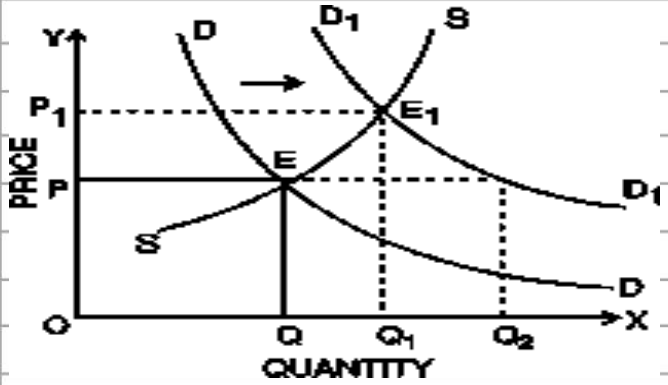
(a) An increase (shift to the right) in demand;

(b) A decrease (shift to the left) in demand;

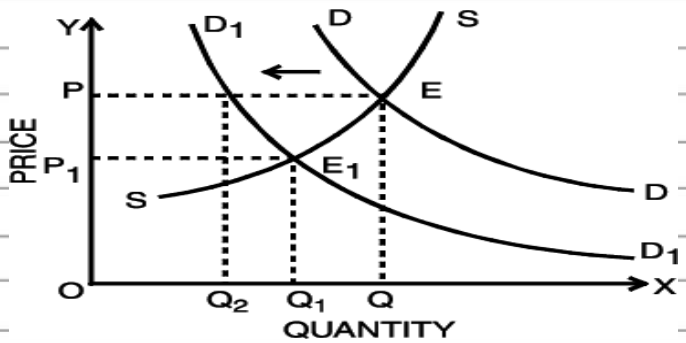
(c) An increase (shift to the right) in supply;

(d) A decrease (shift to the left) in supply.

**(a) Effect on 'P' & 'Q' on increase / decrease in demand (i.e. right / left shift of demand curve)**



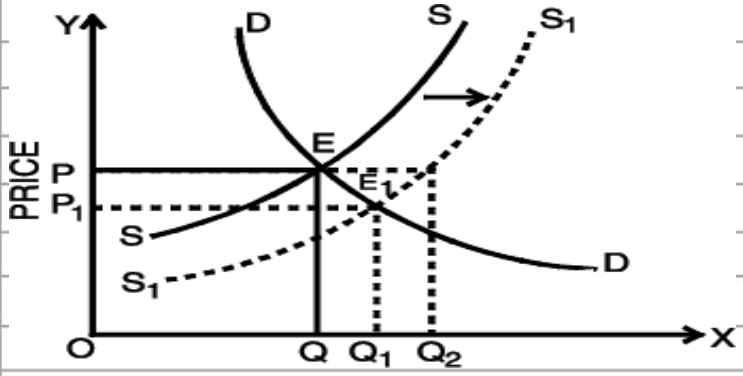

**Increase in Demand**



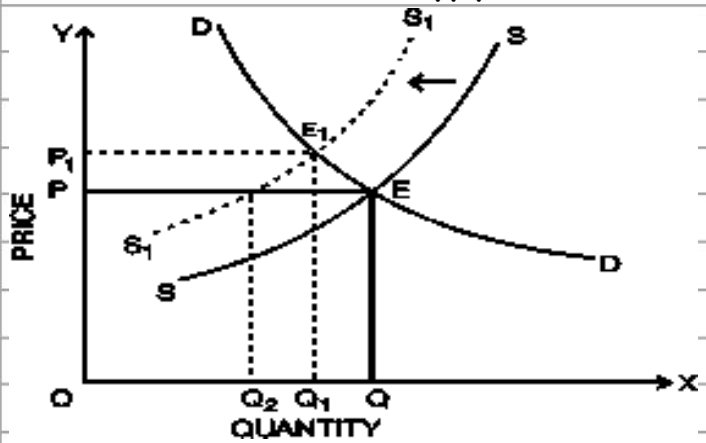

**Change in Demand**

**Conclusion:** D ↑ → P ↑ Q ↑ ; D ↓ → P ↓ Q ↓

(b) Effect on 'P' and 'Q' on increase / decrease in supply (i.e. right/ left shift of SC)



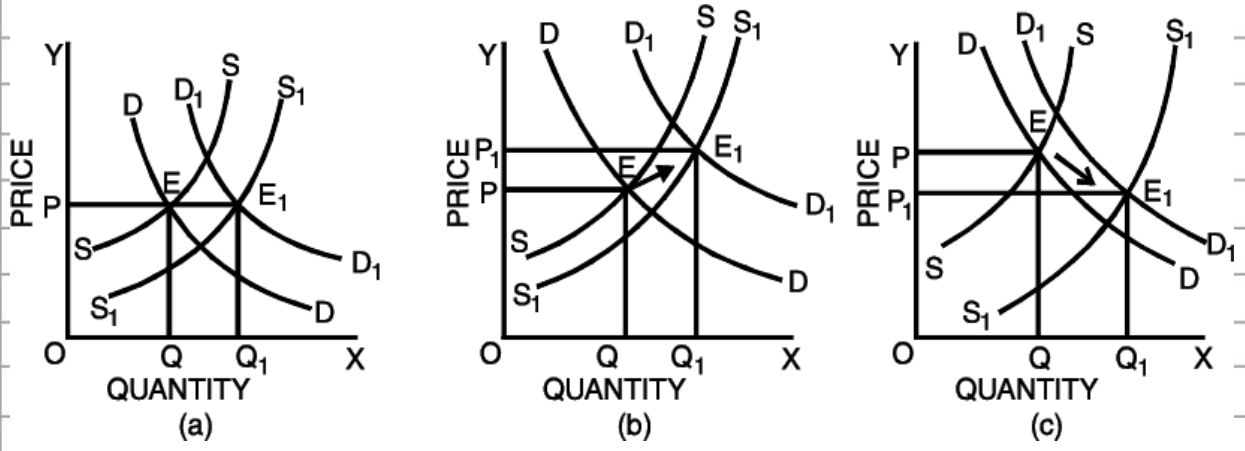

Increase in Supply




Change in Supply

**Conclusion:** ↑S : P ↓ Q ↑ ; S ↓ : P ↑ Q ↓

(c) Effect on 'P' & 'Q' on increase/ decrease in demand and supply simultaneously



**Conclusion:** If demand & supply both are increasing then we can not determine the effect on price but the 'Q' will increase. Similarly if demand and supply both are decreasing then we can not determine the effect on price but the 'Q' will decrease.

## Overall Conclusion of this chapter

| S. No. | Change in Demand/ supply | Change in Equilibrium price | Change in Equilibrium quantity | Remark |
|---|---|---|---|---|
| 1 | D ↑ | P ↑ | Q ↑ | |
| 2 | D ↓ | P ↓ | Q ↓ | |
| 3 | S ↑ | P ↓ | Q ↑ | |
| 4 | S ↓ | P ↑ | Q ↓ | |
| 5 | D ↑ S ↑ | | Q ↑ | |
| 6 | D ↑ 20% S ↑ 20% | P = | Q ↑ | |
| 7 | D ↑ 20% S ↑ 10% | P ↑ | Q ↑ | |
| 8 | D ↑ 10% S ↑ 20% | P ↓ | Q ↑ | |
| 9 | D ↓ S ↓ | | Q ↓ | |
| 10 | D ↓ 20% S ↓ 20% | P = | Q ↓ | |
| 11 | D ↓ 20% S ↓ 10% | P ↓ | Q ↓ | |
| 12 | D ↓ 10% S ↓ 20% | P ↑ | Q ↓ | |
| 13 | D S ↓ | P ↑ | | |
| 14 | D ↓ S ↑ | P ↓ | | |

# CHAPTER – 4 : PRICE DETERMINATION IN DIFFERENT MARKETS

## UNIT III: PRICE OUTPUT DETERMINATION UNDER DIFFERENT MARKET FORMS

### 3.0 PERFECT COMPETITION MARKET

**I. FEATURES**

(1) **Large number of buyer and seller:** Due to this share of each buyer and seller in total demand & supply respectively are "Very small", hence they are not in position to influence the market price through their change in demand & supply.

(2) **The product is identical / homogeneous:** Due to this all goods must be sell at a single market price, hence all firm in this market have to adopt such market price so they are called "Price taker firm".

(3) **Free entry / exit:** New firm can enter to the market without any legal or other barrier and also existing firm can exit the market. Due to above free entry / exit all firms in the market in long-run can not earn 'Super normal profit' or can not bear "Loss" but they can earn only NORMAL PROFIT.

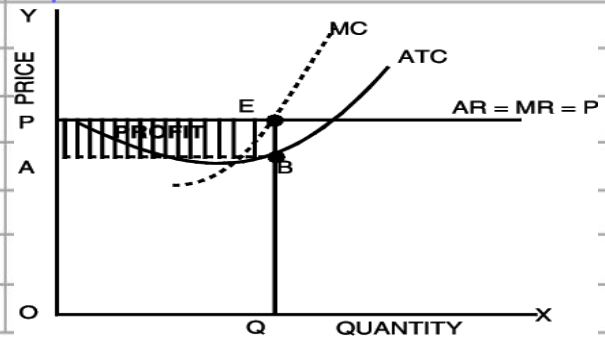
(4) **Perfect knowledge of market condition to both buyer and seller.**

(5) **Very low transaction and transportation cost.**

**Note:** If any firm fulfill first three condition then it will be called "Pure competition" and if fulfill all conditions then it will be "Perfect competition".

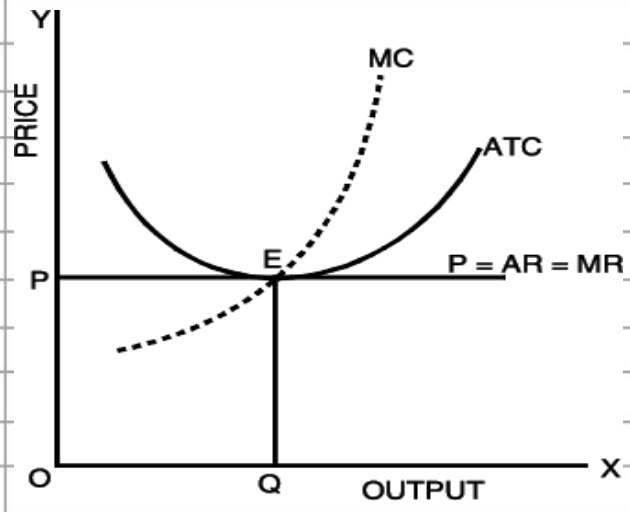
**II. PRICE OUTPUT DETERMINATION OF A FIRM IN PERFECT COMPETITION MARKET:**

**(A) Super Normal Profit**



**Conclusion:** If AR > AC at equilibrium level of output then we generate super normal profit (SNP) and area is A,B,E,P.

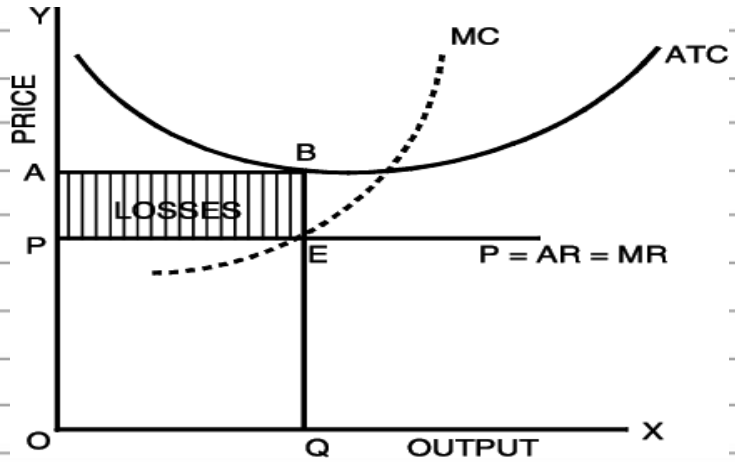
## (B) Normal Profit




**Conclusion:** AR = AC at equilibrium level of output therefore PC-firm is earning just normal profit.

Logic- Why we are calling it as normal profit where AR = AC (as per normal concept this is a situation of no profit no loss [BEP]) but in economics AC also includes normal profit of owners.

## (C) Loss



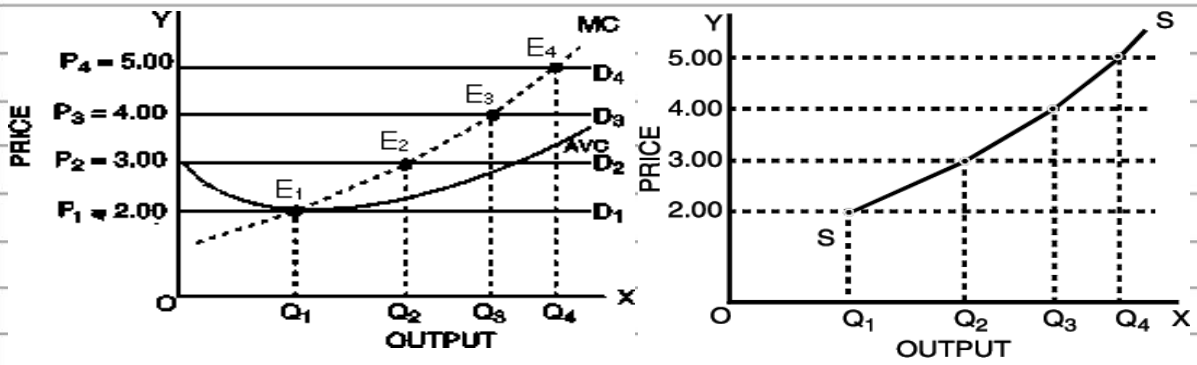

**Final conclusion:** At equilibrium level of output, if

AR > AC then firm in earning SNP

AR = AC then firm in earning NP

AR < AC then firm in earning LOSS

## III. SHORT RUN SUPPLY CURVE OF THE PC-FIRM




**Conclusion**

(1) In PC firm, the 'MC curve above AVC' is identical shape of its 'supply curve'.

(2) As per above diagram the supply curve will be point $E_1$, $E_2$, $E_3$, $E_4$ & f and onwards but it can not be point '$E_0$' in MC curve, because at point '$E_0$' the price (AR) is $P_0$ which is less the AVC, therefore as per P-1 firm should not produce / supply if AR < AVC.

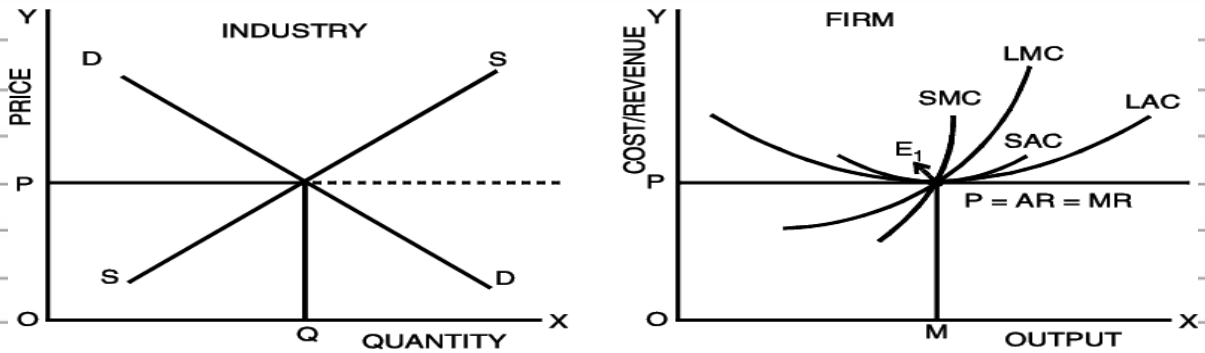
## IV. LONG RUN EQUILIBRIUM OF PC-FIRM

(1) In PC market due free entry / exit, the firm can only earn normal profit in long run.

(2) If in short run the firm is earning SNP, then new firms will enter to the market in long run, therefore the supply of market will increase and price of market will decrease therefore the firm have to take this reduced price and their AR (i.e. price) will be reduced to AC and the firm can earn only NP.

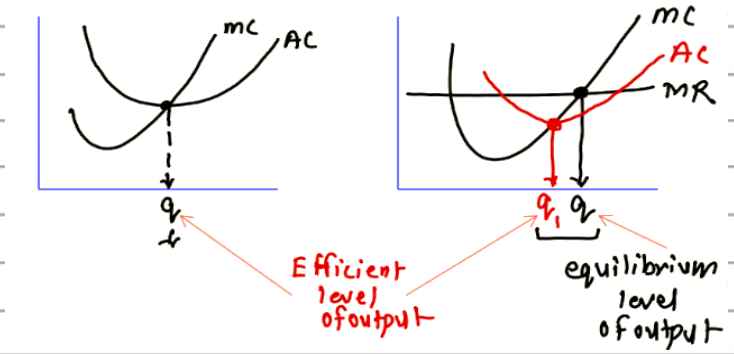
(3) If in short run the firms having loss, the existing firm will exit the market and supply of market will decrease, therefore price of market will increase and all the firm can take such increased price so AR increases upto AC and earn NP.

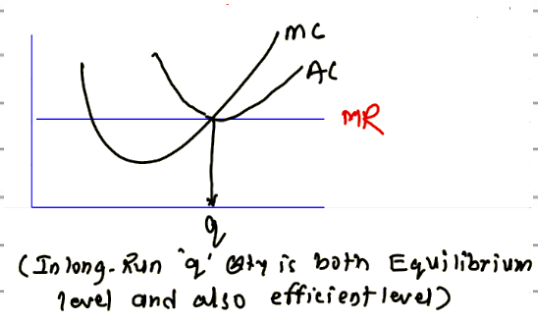
(4)



**Note:**

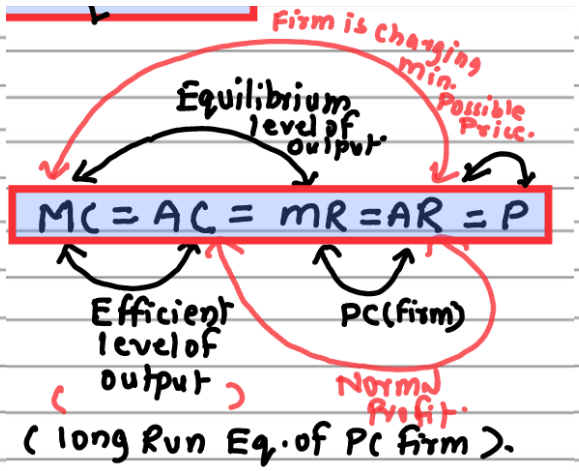
(1) In long-run all firm in PC market will adjust its capacity so that their cost (AC) is minimum (i.e. the firm will produce an output where equilibrium level is just equal to efficient level of output).




(2) At long-run equilibrium

Equilibrium output




## V. Long run equilibrium of industry

The PC market (Industry) will be in equilibrium if following conditions are fulfill:

(1) All firms are in equilibrium

(2) No firm has an incentive to entry / exit.

(3) Market demand = market supply (i.e. no surplus/ shoftage)

In Equilibrium of Industry "Market Mechanism" leads to optimal allocation of resources and following effect will be seen:

(a) All firms will produce output at minimum cost (i.e. AC = MC)
(b) All firms will sale at minimum possible price (i.e. MC = AR)
(c) All firm is using its full capacity and no wastage of resources (MC = AC)
(d) All firms are earnings just normal profit (i.e. AR = AC)
(e) All firms are maximizing its profit (i.e. MR = MC)
(f) There is optimum number of firm in industry.

'PC' Market is a "Myth"

## 3.1 MONOPOLY

### I. Features

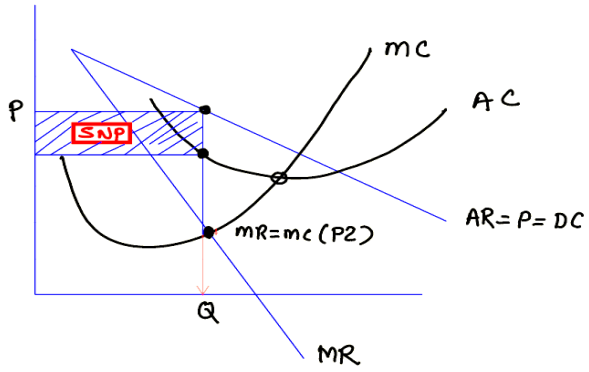
(1) Single seller and many buyer.
(2) Unique goods (i.e. no close substitute) $E_c = 0$.
(3) Entry barriers: Economical, legal etc.
(4) Market power: More control over price i.e. $E_p < 1$.

### II. HOW DO MONOPOLY ARISE?

1. Control over raw material, technology etc.
2. Unique product (i.e. difficult to produce / costly to produce)
3. Patents and copyrights
4. Cartels (illegal in most countries)
5. Extremely large setup / start-up (ex. Railways)
6. Natural monopoly: if total demand of industry is required for minimum cost (efficient level of output) and it is producing by a single firm.
7. goodwill: goodwill of a single firm create entry barriers.
8. High legal requirement to start business.
9. Predatory tactics: Existing firm reduce prices when new firm want to entry.
10. Exclusive right provided by government to produce & sell.

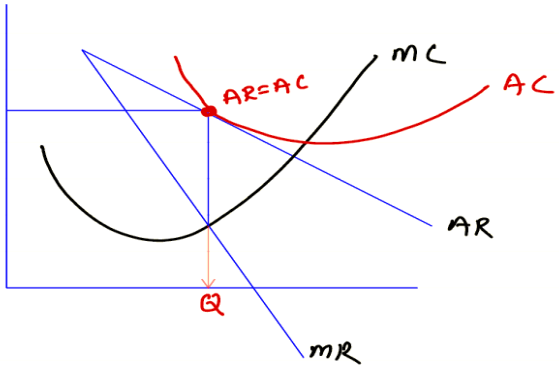
## III. PRICE - OUTPUT DETERMINATION IN A MONOPOLY MARKET:

### (a) SNP: When AR > AC at equilibrium output

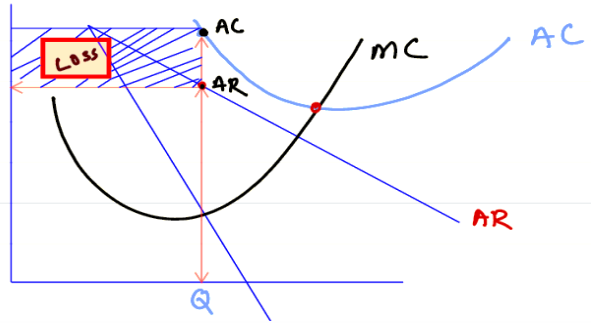



**Supernormal profits of a competitive firm**

### (b) NP: AR = AC at equilibrium output




### (c) LOSS at equilibrium output




## IV. LONG-RUN EQUILIBRIUM OF MONOPOLY

(1) Firm may adjust its plant capacity for achieve efficient level of output or may be use existing capacity in long run also. The firm can bear high cost due to no competition.

(2) Due to point (1) above the firm may produce at sub efficient level / sub-optimal level and it will not be at minimum point of LAC.

(3) Firm may be in super normal profit in long run because the entry of new firm is blocked.

(4) If there is loss in long run, the firm should close its business.

## V. PRICE DISCRIMINATION BY MONOPOLIST:

**(a) Meaning:** "It means selling a product for different prices"

**Example:** Following business generally have price discrimination

(1) Electricity company

(2) Railway

(3) ICAI journals

**(b) Conditions of price discrimination:**

(1) Firm must have "Price making power".

(2) Firm can divide its market into two or more sub-market.

(3) Different price elasticity in different market.

(Ep ↓ P ↑ Vs Ep ↑ P ↓)

(4) It should not be possible for buyer of low price market to resell the product to the buyer of high price market. (i.e. No market arbitrage).

**(c) Objective of price discrimination**

(1) Earn maximum profit.

(2) Dispose off surplus stock.

(3) Enjoy economies of scale.

(4) Capture foreign market.

(5) Secure equity through pricing.

### Formula of marginal revenue (MR)

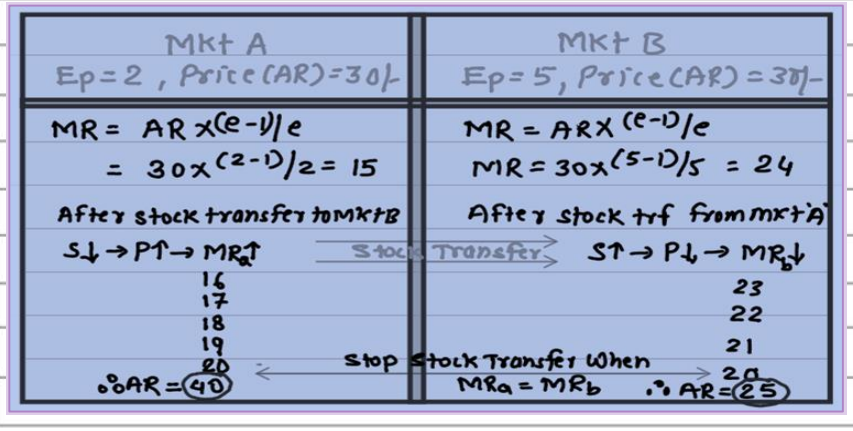
(a) $MR_n = TR_n - TR_{n-1}$

(b) $MR = \Delta TR / \Delta Q$

or

(c) $MR = AR \times \left(\frac{e-1}{e}\right)$

**Example:**



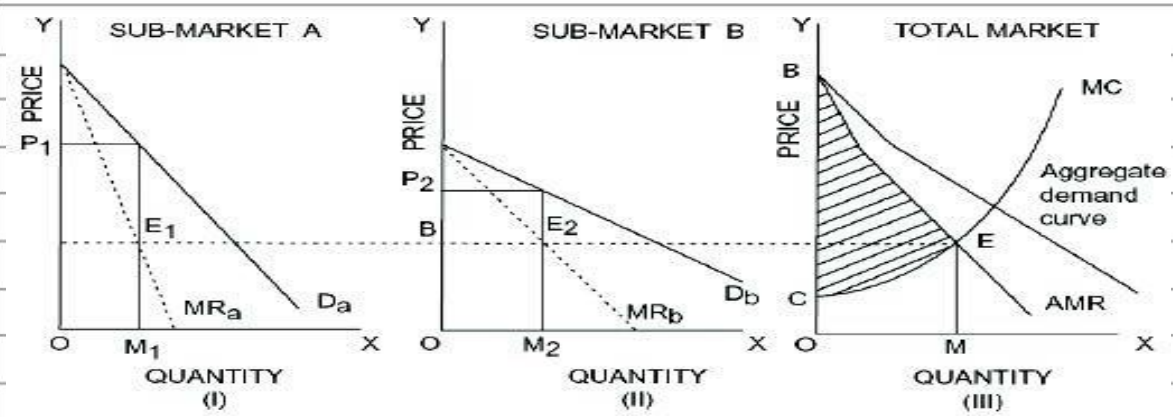
**Process:**

(1) Monopolist transfer stock from market A to Market B, then price of Market A start increasing and price of market B start decreasing due to decrease / increase in supply.

(2) Due to point (1) above $MR_a$ start increasing and $MR_b$ start decreasing.

(3) Monopolist transfer stock until $MR_a = MR_b$.

(4) The final price of Market A & Market B will be at $MR_a = MR_b$.




**Fixation of total output and price in the two sub-markets by the discriminating monopolist**

**As per "Pro. Pigou" price discrimination classified in to three degrees:**

(a) **First degree:** Charge a price just equal to a price which consumer is ready to pay i.e. MU. In this degree consumer surplus will be ZERO.

(b) **Second degree:** Charge below the price a consumer is ready to pay (i.e. P < MU), therefore part of consumer surplus left with consumer.

(c) **Third degree:** discriminate price as per difference market, buyer, location, customer segment etc.

## VI. ECONOMIC EFFECT OF MONOPOLY

(1) Loss of productive efficiency

(2) Higher price and low production

(3) Super normal profit even in long run

| | |
|---|---|
| (4) | Consumer is not able to substitute goods & there is a reduction in consumer surplus. |
| (5) | Restrict 'consumer sovereignty' |
| (6) | Barriers of entry of new firm |
| (7) | Create political power in order to obtain favorable legislation (law) |
| (8) | No innovations |
| (9) | Monopoly firm pay less price for inputs also |
| (10) | 'X' inefficiency. |

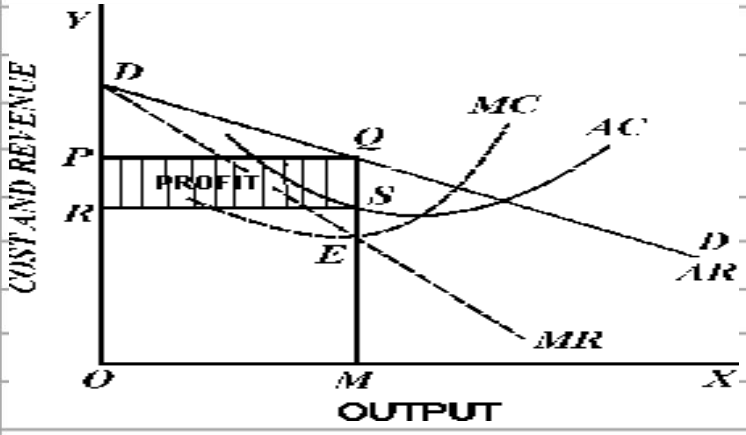
## 3.2 MONOPOLISTIC COMPETITION

### 3.2.0 FEATURES MONOPOLISTIC COMPETITION

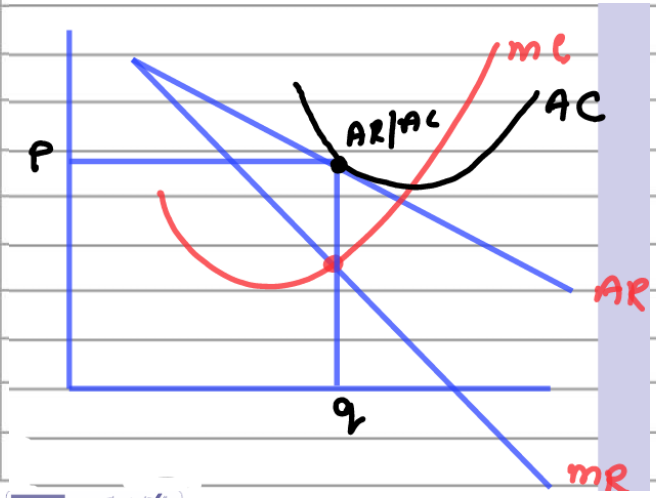
| | |
|---|---|
| (i) | Large numbers of sellers |
| (ii) | Product differentiation |
| (iii) | Free entry / exit |
| (iv) | Non -price competition |

### 3.2.1 PRICE OUTPUT DETERMINATION

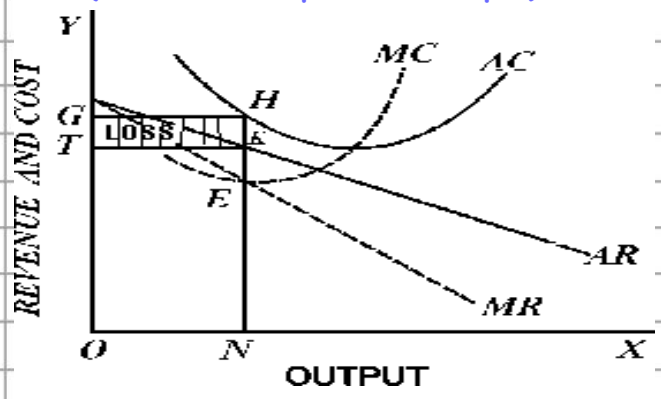
(a) Super normal profit (when AR > AC at equilibrium output)




(b) NP (when AR = AC at equilibrium output)



(c) Loss (AR < AC at equilibrium output)




Long run equilibrium in monopolistic competition

(1) Due to free entry / exit the firm will be in "Normal profit".

(2) The firm maintain excess capacity in long run, therefore it will not be in minimum cost.

(3) Excess capacity is maintain to fulfill sudden demand and increase market share.

## 3.3 OLIGOPOLY:

### I. FEATURES

(1) Few seller (two to ten), many buyer and deal will any product.

(2) "A situation in which a firm bases its market policy, in part, on the expected behavior of a few close rivals" – Prof. Stigler.

(3) Oligopoly arises due to those factors which are responsible for creating monopoly.

(4) Example of oligopoly market

(a) Automobile industry

(b) Petroleum refining

(c) Power generation

(d) Telecommunication

(e) Airlines etc.

**II. TYPES OF OLIGOPOLY**

(a) **Pure Oligopoly or perfect oligopoly:** If product is homogeneous then its pure oligopoly.

(b) **Open and close oligopoly:** If free entry for new firm then its open oligopoly otherwise close oligopoly.

(c) **Collusive and competitive oligopoly:** When there is common understanding / collusion amongst oligopolist, then it is called collusive oligopoly, otherwise competitive oligopoly.

(d) **Partial and full oligopoly:** If one large firm dominate the other firms then it will be partial oligopoly. (Ex. LIC in insurance business). If all firm have equal share then its full oligopoly.

(e) **Syndicated and organized oligopoly:** If all firm sell their product through a centralized market/ syndicate, it will be syndicated oligopoly. Organized oligopoly refer to a situation where firm organized themselves into a central association for fixing price, output, quotas etc.

## 3.3.0 CHARACTERISTICS OF OLIGOPOLY MARKET

(1) **Strategic interdependence**

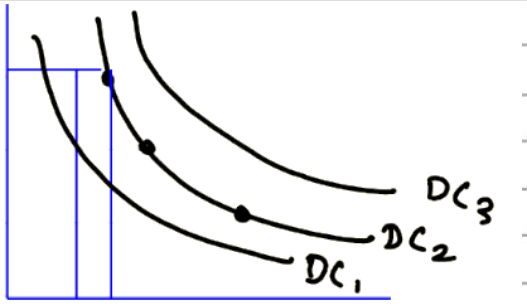
(2) **Importance of advertising and selling cost**

(3) **Group behavior**

## 3.3.1 PRICE – OUTPUT DECISIONS IN OLIGOPOLY MARKET

(1) Its price policy is depend on its rivals, if behavior of rivals are less competitive, then the firm enjoy "market power" and they can charge price above "marginal cost".

(2) Oligopoly firm can not have sure or determinate "demand curve".



## 3.3.2 PRICE LEADERSHIP:

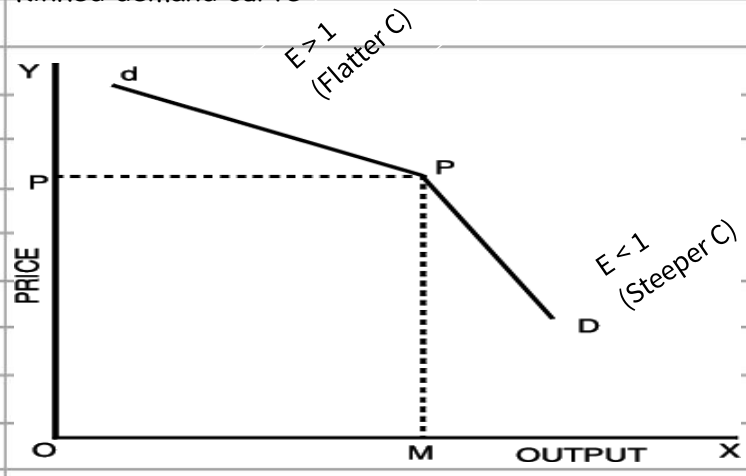
**As Firm Is Not Small.**

## 3.3.3 KINKED DEMAND CURVE

(a) Normally in this industry price remain same/ sticky/ inflexible for long time (price rigidity).

(b) Price rigidity is explained by American economist Paul A. Sweezy (Sweezy's model)

(c) Kinked demand curve




(i) The 'Demand Curve' will be Kinked at market price (i.e. 'P')

(ii) When any firm want to increase price then its rivals does not increase price so firm Ep > 1 so firm don't want to increase price.

(iii) When any firm decrease price then its rivals also decrease price and firm could not increase demand due to Ep < 1.

(iv) Due to (ii) & (iii) above the price in this market will same for long time.

## 3.3.4 OTHER IMPORTANT FORM

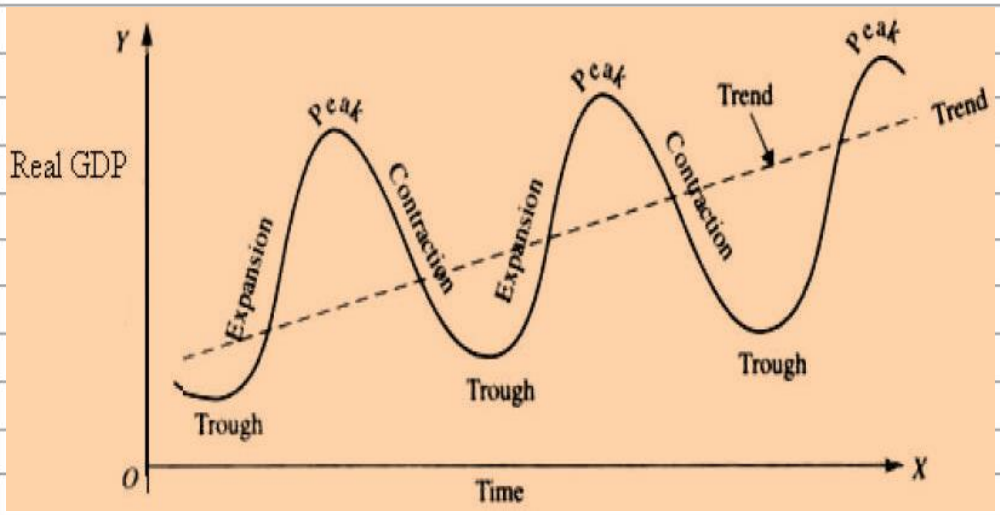
(a) **Duopoly:** If only two firm in the market.

(b) **Monopsony:** only single buyer in the market (Ex. Market of goods only used by railway)

(c) **Oligopsony:** only few buyer in the market (Ex. Market of Raw material only used by oligopoly market)

(d) **Bilateral monopoly:** only single seller and single buyer. (Ex. Single producer of goods purchased only by Railway).

# CHAPTER – 5: BUSINESS CYCLES




## 5.0 Introduction

**Case 1:** Rapid growth of UK during 1920s: Due to "Assembly line".

**Case 2:** China's economic slowdown: China's slow down effect all countries.

## 5.1 Phases of business cycle

(1) Expansion (upswing) ⟶ Demand ↑

(2) Peak

(3) Contraction ⟶ Demand ↓

(4) Trough

**Economic Indicators of turning point**

(1) **Leading indicators:** This indicators changes before the turning point. For example: change in share/ stock price, new orders of machinery / capital goods etc.

(2) **Coincident / concurrent indicator:** This indicators changes simultaneously with the turning point. For example: GDP, Inflation, Industrial production, retail sales etc.

(3) **Lagging indicators:** This indicators changes only after the turning point. For example: unemployment, corporate profit, labour cost per unit.

**Example of business cycle:-**

(a) **Great depression of 1930:** The world economy suffer the longest, deepest and the most widespread depression of 20th century during 1930. It started in the "US" and become worldwide. The reason of this depression is (i) as per maynard Keynes is lower aggregate expenditure (ii) as per monetarists it is due to banking crisis and low money supply. (iii) as per some economists, it is due to deflation, over- in debtness, lower profit etc.

(b) **Information technology bubble burst of 2000:** It may be called Dot.com bubble bust, during 1997 to 2000 due to low interest rate and over optimism the share price of IT co. start increasing and bank also funding more & more due to this company offer their service for free with expection of future profit. Those company could not sustain for long and the collapse of bubble took place during 1999-2000.

(c) **Global Economic crisis (2008 – 09):** It start from 'US' financial market. 'US' central bank reducer rate of interest therefore the 'US' banks start over funding for housing loan even without checking the paying capacity. Due to this supply of houses are increasing as household began to buy houses in increasing numbers therefore borrower start defaulting the loan (NPA). Then housing bubble got bust during 2007. Therefore loss to the bank and due to the chain effect whole US economy and world economy felt its impact.

## 5.2 Features of Business Cycles

(a) Business cycles occur periodically although they do not exhibit the same regularity. The duration of these cycles vary. The intensity of fluctuations also varies.

(b) Business cycles have distinct phases of expansion, peak, contraction and trough. These phases seldom display smoothness and regularity. The length of each phase is also not definite.

(c) Business cycles generally originate in free market economies. They are pervasive as well. Disturbances in one or more sectors get easily transmitted to all other sectors.

| | |
|---|---|
| **(d)** | Although all sectors are adversely affected by business cycles, some sectors such as capital goods industries, durable consumer goods industry etc, are disproportionately affected. Moreover, compared to agricultural sector, the industrials sector is more prone to the adverse effects of trade cycles. |
| | |
| **(e)** | Business cycles are exceedingly complex phenomena; they do not have uniform characteristics and causes. They are caused by varying factors. Therefore, it is difficult to make an accurate prediction of trade cycles before their occurrence. |
| | |
| **(f)** | Repercussions of business cycles get simultaneously felt on nearly all economic variables viz. output, employment, investment, consumption, interest, trade and price levels. |
| | |
| **(g)** | Business cycles are contagious and are international in character. They begin in one country and mostly spread to other countries through trade relations. For example, the great depression of 1930s in the USA and Great Britain affected almost all the countries, especially the capitalist countries of the world. |
| | |
| **(h)** | Business cycles have serious consequences on the well-being of the society. |

## 5.3 Causes of Business Cycles

| | |
|---|---|
| **(i)** | **Internal Causes:** |
| (a) | Fluctuations in Effective Demand |
| (b) | Fluctuations in Investment |
| (c) | Variations in government spending |
| (d) | Macroeconomic policies |
| (e) | Money Supply |
| (f) | Psychological factors |
| | |
| **(ii)** | **External Causes** |
| (a) | Wars |
| (b) | Post War Reconstruction |
| (c) | Technology Shocks |
| (d) | Natural Factors |
| (e) | Population growth |

# GLOSSARY

1. **Economics:** It the branch of knowledge which is concerned with production, consumption and transfer of wealth.
2. **Business Economics:** The use of economic analysis to make business decisions involving the best use of an organization's scarce resources.
3. **Micro Economics:** It is basically the study of behaviour of different individuals and organizations within an economic system. Here the focus is on a small number of group of units rather than all the units combined.
4. **Macro Economics:** It is the study of the overall economic phenomena of the economy as a whole, rather than its individual parts. Accordingly, in Macro Economics, we study the behaviour of the large economic aggregates.
5. **Positive Economics:** It is the branch of economics that concerns the description and explanation of economic phenomena. It focuses on facts and cause and effect relationships.
6. **Normative Economics:** It is that part of economics that expresses value judgements about economic fairness or what the outcome of the economy or goals of public policy ought to be.
7. **Economic System:** An economic system refers to the sum total of arrangements for the production and distribution of goods and services in a society. It includes various individuals and economic systems.
8. **Capitalist Economy:** An economic system in which all means of production are owned and controlled by private individuals for profit.
9. **Socialist Economy:** An economic system where the resources are allocated according to the commands of a central planning authority and market forces have no role to play in the allocation of resources.
10. **Mixed Economy:** An economic system which depends on both markets and governments for allocation of resources. The aim is to include the best features of both Capitalist and Socialist Economy.
11. **Demand:** The various quantities of a given commodity or service which the consumers would buy in one market during a given period of time, at various prices, or at various incomes, or at various prices of related goods.

| | |
|---|---|
| 12. | **Market Demand:** It is defined as the sum of individual demands for a product at a price per unit of time. |
| 13. | **Elasticity of Demand:** It is defined as the responsiveness of the quantity demanded of a good to changes in one of the variables on which demand depends. More precisely, elasticity of demand is the percentage change in quantity demanded divided by the percentage change in one of the variables on which demand depends. |
| 14. | **Inferior Goods:** Inferior goods are those goods whose quantity demanded decreases with the increase in money income. |
| 15. | **Price Elasticity:** It expresses the response of quantity demanded of a good to a change in its price, given the consumer's income, his tastes and prices of all other goods. |
| 16. | **Income Elasticity:** It is the degree of responsiveness of quantity demanded of a good to changes in the income of consumers. |
| 17. | **Cross Demand:** It refers to the quantities of a commodity or service which will be purchased with reference to changes in price, not of that particular commodity, but of other inter-related commodities, other things remaining the same. |
| 18. | **Cross Elasticity:** A change in the demand for one good in response to a change in the price of another good represents cross elasticity of demand of the former good for the latter good. |
| 19. | **Advertisement Elasticity:** Advertisement elasticity of sales or promotional elasticity of demand is the responsiveness of a good's demand to changes in firm's spending or advertising. The advertising elasticity of demand measures the percentage change in demand that occurs given a one percent change in advertisement expenditure. |
| 20. | **Producers Goods:** Producers goods are those goods which are used for the production of other goods- either consumer goods or producer goods themselves. |
| 21. | **Consumer Goods:** Those goods which are used for final consumption. |
| 22. | **Utility:** Utility is the anticipated satisfaction by the consumer, and satisfaction is the actual satisfaction derived. |
| 23. | **Total Utility:** It is the sum of utility derived from different units of a commodity consumed by a consumer. |
| 24. | **Marginal Utility:** It is the addition made to total utility by the consumption of an additional unit of a commodity. |

| | |
|---|---|
| 25. | **Consumer Surplus:** Is is defined as the excess of price that the the consumer is ready to pay from which he actually pays. |
| 26. | **Indifference Curve:** It is a curve which represents all those combinations of two goods which give same satisfaction to the consumer. |
| 27. | **Indifference Map:** A collection of many indifference curves where each curve represents a certain level of satisfaction. In short, a set of indifference curve is called indifference map. |
| 28. | **Supply:** It refers to the amount of a good or service that the producers are willing and able to offer to the market at various prices during a given period of time. |
| 29. | **Elasticity of Supply:** It is defined as the responsiveness of the quantity supplied of a good to a change in its price. |
| 30. | **Equilibrium Price:** The price at which the wishes of both the buyers and sellers are satisfied. At this price, the amount that buyers want to buy and sellers want to sell are equal. |
| 31. | **Production:** It is the organized activity of transforming resources into finished products in the form of goods and services. |
| 32. | **Land:** The term "land" is used in a special sense in economics. It does not mean soil or earth's surface alone, but refers to all free gifts of nature. |
| 33. | **Labour:** The term labour means any mental or physical exertion directed to produce goods and services. |
| 34. | **Capital:** Capital is that part of wealth of an individual which is used for further production of wealth. Capital is a stock concept which yields a periodical income which is a flow concept. |
| 35. | **Entrepreneur:** A factor which mobilizes all other factor of production like land, labour, capital, and combines them in the right proportion, initiates the process of production and bears the risk involved in it. |
| 36. | **Average Product:** Average product is the total product per unit of the variable factor. |
| 37. | **Marginal Product:** It is the change in total product per unit change in the quantity of variable factor. |
| 38. | **Isoquant:** An isoquant represents all those combinations of inputs which are capable of producing the same level of output. |
| 39. | **Cost Analysis:** The study of behaviour of cost in relation to one or more production criteria, namely, size of output, scale of operations, prices of the factor of production and other relevant economic variables. |

40. **Accounting Costs:** Accounting costs relate to those costs which involve cash payments by the entrepreneur of the firm. These are explicit cost & the expenses already incurred by the firm.

41. **Economic cost:** The cost which takes into account the explicit as well as the implicit cost is known as Economic cost.

42. **Social Cost:** Social cost refers to the total cost borne by the society on account of a business activity and includes private cost and external cost.

43. **Fixed Cost:** The costs which do not vary with the level of output upto a certain level of activity are known as fixed cost.

44. **Variable Cost:** These costs are a function of output and hence vary with the production.

45. **Marginal Cost:** Marginal cost is the addition made to the total cost by production of an additional unit of output.

46. **Market:** A market is a collection of buyers and sellers with a potential to trade. The actual or potential interactions of the buyers and sellers determine the price of a product or service.

47. **Perfect competition:** It is a type of market which is characterized by many sellers selling identical products to many buyers.

48. **Monopoly:** It is a situation where there is a single seller and many buyers. The product sold does not have any close substitutes.

49. **Monopolistic competition:** This type of market is characterized by many sellers selling differentiated products to many buyers.

50. **Oligopoly:** There a few sellers selling competing products to many buyers.

51. **Average revenue:** It is the revenue earned per unit of output. It is nothing but the price of one unit of output.

52. **Marginal revenue:** It is the change in total revenue resulting from the sale of an additional unit of the commodity.

53. **Price discrimination:** It is a method of pricing adopted by a monopolist to earn abnormal profits. It refers to the practice of charging different prices for different units of the same commodity.

54. **Cartel:** Cartel refers to a group of firms that explicitly agree to coordinate their activities.

55. **Business Cycle:-** The rhythmic fluctuations in aggregate economic activity that an economy experiences over a period of time are called business cycles or trade cycles. A typical business cycle has four distinct phases namely Expansion, Boom, Contraction, Trough.